MATH 5080 Class Notes

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1 The real number system

1.1 Natural numbers, integers, and rationals

Definition 1.1 (Natural numbers)**.** We define the set $\{1, 2, 3, 4, \ldots\}$ as the set of **natural numbers**, denoted $\mathbb{N}$.

Remark. $\mathbb{N}$ is closed under addition and multiplication, but not subtraction.

Definition 1.2 (Integers)**.** We define the set $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$ to be the set of **integers** denoted $\mathbb{Z}$.

Remark. Integers are closed under subtraction, but not division.

Definition 1.3 (Rationals)**.** We define the set

$$\left\{\frac{p}{q} : p, q \in \mathbb{Z}, q \neq 0\right\}$$

to be the set of **rational numbers**, denoted $\mathbb{Q}$.

Remark. $\mathbb{Q}$ has holes. For example, it can be shown that $x^2 = 2$ does not have a solution in $\mathbb{Q}$.

Proof. Suppose otherwise, then

$$\exists p, q \in \mathbb{Q}, q \neq 0, s.t. \frac{p^2}{q^2} = 2$$

WLOG, we can pick p, q such that they are coprime i.e. $gcd(p, q) = 1$.

$$\begin{aligned}
\frac{p^2}{q^2} = 2 &\implies p^2 = 2q^2 \\
&\implies p = 2k \text{ for sume } k \\
&\implies (2k)^2 = 2q^2 \\
&\implies 2k^2 = q^2 \\
&\implies q = 2k' \text{ for some } k' \\
&\implies gcd(p, q) \geq 2 > 1
\end{aligned}$$

✓

Remark. $x^2 = p$ for any prime p does not have a solution in $\mathbb{Q}$.

Proof. Assume otherwise, then

$$\begin{aligned}
&\exists m, n \in \mathbb{Z}, n \neq 0, s.t. \frac{m^2}{n^2} = p, gcd(m, n) = 1 \\
\implies & m^2 = n^2 p \\
\implies & m = pk \text{ for some } k, \text{ since } p \text{ is prime} \\
\implies & p^2 k^2 = n^2 p \\
\implies & pk^2 = n^2 \\
\implies & n = pk' \text{ for some } k' \text{ since } p \text{ is prime} \\
\implies & gcd(m, n) \geq p > 1
\end{aligned}$$

✓

Definition 1.4 (Real numbers)**.** The **real numbers**, denoted $\mathbb{R}$ is a set that

(i) is an ordered field
(ii) contains $\mathbb{Q}$ as an sub-ordered field
(iii) satisfies the completness axiom, or the least upper bound property

Remark. Such a set is unique under isomorphisms.

1.2 Ordered Field

Definition 1.5 (Field)**.** : A field F is a set with two binary operations

$$+ : F \times F \to F,\ (x, y) \mapsto x + y$$

$$\cdot : F \times F \to F,\ (x, y) \mapsto x \cdot y$$

that satisfy these properties:

- (A1) additive closure: $x, y \in F, x + y \in F$
- (A2) additive commutativity: for all $x, y \in F$, $x + y = y + x$
- (A3) additive associativity: for all $x, y, z \in F$, $x + (y + z) = (x + y) + z$
- (A4) existence of additive identity or neutral element: there is $0 \in F$ such that $x + 0 = x$ for all $x \in F$
- (A5) existence of additive inverse: for all $x \in F$ there is y such that $x + y = 0$
- (M1) multiplicative closure: $x, y \in F, x \cdot y \in F$
- (M2) multiplicative commutativity: for all $x, y \in F$, $x \cdot y = y \cdot x$
- (M3) multiplicative associativity: for all $x, y, z \in F$, $x \cdot (y \cdot z) = (x \cdot y) \cdot z$
- (M4) existence of multiplicative identity or neutral element: there is $1 \in F, 1 \neq 0$ such that $x \cdot 1 = 1 \cdot x = x$ for all x
- (M5) existence of multiplicative inverse: for all $x \in F, x \neq 0$ there is y such that $x \cdot y = 1$
- (D) distributivity: for all $x, y, z \in F$, $(x + y) \cdot z = x \cdot z + y \cdot z$

Remark. Additive neutral element, 0, is unique.

Proof. Suppose otherwise, i.e. $\exists 0, 0' \in F$ both satisfying A4, then

$$\begin{aligned} 0 &= 0 + 0' \text{ by A4, with } 0' \text{ as identity} \\ &= 0' + 0 \text{ by commutativity} \\ &= 0' \text{ by A4, since 0 is additive identity} \end{aligned}$$

✓

Remark. Multiplicative identity, 1, is unique.

Proof. Suppose otherwise, i.e. $\exists 1, 1' \in F$ both satisfying A4, then

$$\begin{aligned} 1 &= 1 \cdot 1' \text{ by M4, with } 1' \text{ as identity} \\ &= 1' \cdot 1 \text{ by commutativity} \\ &= 1' \text{ by M4, since 1 is identity} \end{aligned}$$

✓

Remark. $\forall x \in F$, the additive inverse is unique.

Proof. Suppose otherwise, then $\exists y, y'$ s.t. $x + y = x + y' = 0$.

$$\begin{aligned} y &= y + 0 \\ &= y + (x + y') \\ &= (y + x) + y' \text{ by associativity} \\ &= (x + y) + y' \text{ by commutativity} \\ &= y' \end{aligned}$$

✓

Remark. $\forall x \in F, x \neq 0$, the multiplicative inverse is unique.

Proof. Suppose otherwise, then $\exists y, y'$ s.t. $x \cdot y = x \cdot y' = 1$.

$$\begin{aligned} y &= y \cdot 1 \\ &= y \cdot (x \cdot y') \\ &= (y \cdot x) \cdot y' \text{ by associativity} \\ &= (x \cdot y) \cdot y' \text{ by commutativity} \\ &= y' \end{aligned}$$

✓

Remark. Since the additive and multiplicative inverses are unique, we denote $-x$ and $\frac{1}{x}$, and we define subtraction as

$$x - y := x + (-y)$$

We define division as

$$\frac{x}{y} := x \cdot \left(\frac{1}{y}\right), y \neq 0$$

Proposition 1.6. Let $x, y, z \in F$, then

1. $x + y = y + z \implies y = z$
2. $-(-x) = x$
3. $xy = xz, x \neq 0 \implies y = z$
4. $\frac{1}{1/x} = x$ if $x \neq 0$
5. $0 \cdot x = 0$
6. $-0 = 0$
7. $(-1) \cdot x = -x$
8. $xy = 0 \iff x = 0$ or $y = 0$
9. $(-x)(-y) = xy$

Definition 1.7 (Ordered field)**.** A field F is an **ordered field** if it can be equipped with a relation $<$ so that

1. $\forall x, y \in F$, exactly one of $x < y, x = y, y < x$ holds
2. $\forall x, y, z, \in F, x < y, y < z \implies x < z$
3. $\forall x, y, z \in F,\ x < y \implies x + z < y + z$
4. $\forall x, y, \in F,\ 0 < x, 0 < y \implies 0 < xy$

Proposition 1.8. Let F be an ordered field and $x, y, z, w \in F$. Then

1. $x > y, z > 0 \implies xz > yz$
2. $x > y, z < 0 \implies xz < yz$
3. $x > y \iff -x < -y$
4. $x < y, z < w \implies x + z < y + w$
5. $0 < x < y \implies 0 < \frac{1}{y} < \frac{1}{x}$
6. $\forall x \neq 0, x^2 > 0$. In particular, $1 > 0$

Definition 1.9 (Absolute value)**.** For all $x \in \mathbb{R}$, we define its **absolute value** $|x|$ as

$$|x| = \begin{cases} x, & \text{if } x \geq 0, \\ -x, & \text{if } x < 0, \end{cases}$$

Proposition 1.10. Let $x, y \in \mathbb{R}, a \geq 0$, then

1. $|x| \geq 0$
2. $|x| \leq a \iff -a \leq x \leq a$
3. $|xy| = |x| \cdot |y|$
4. (triangle inequality) $|x + y| \leq |x| + |y|$

Proposition 1.11. Let $x, y \in \mathbb{R}$ such that $x \leq y + \epsilon, \forall \epsilon > 0$, then $x \leq y$.

Proof. We prove by contrapositive. Assume that $x < y$, WTS that $\exists \epsilon > 0, y + \epsilon < x$.

Pick

$$\epsilon := \frac{x - y}{2}$$

Then

$$\begin{aligned} y + \epsilon &= y + \frac{x - y}{2} \\ &= \frac{x + y}{2} \text{ by (D), (A2), (A3)} \\ &< \frac{x + x}{2} \text{ by (O3)} \\ &< x \end{aligned}$$

✓

1.3 Completeness Axiom / LUB property

Definition 1.12 (Bounded from above)**.** Let $S \subset \mathbb{R}$, we say S is **bounded from above** if $\exists m \in \mathbb{R}$ s.t. $s \leq m, \forall s \in S$. Such an m is called an **upper bound** of S.

Definition 1.13 (Bounded)**.** Let $S \subset \mathbb{R}$, we say S is **bounded** if it is bounded from both above and below.

Example. $[0, 1] := \{x \in \mathbb{R}, 0 \leq x \leq 1\}$ has upper bounds 1, 1.5, 2, etc.

Example. $\mathbb{N}$ is not bounded from above.

Definition 1.14 (Maximum)**.** If an upper bound m of S is an element of S, then m is called the **maximum** of S, denoted

$$\max S$$

Example. $S = [0, 1\}$ does not have a maximum, i.e. $\max S$ DNE.

Proof. $\forall m \in S$, take $\epsilon = \frac{1}{2}(1 - m)$. Then $m + \epsilon < 1, m + \epsilon > m$. Hence m is not an UB. Hence any m cannot be UB. ✓

Definition 1.15 (Supremum / least upper bound)**.** Let $S \subset \mathbb{R}$ be bounded from above. Then $m \in \mathbb{R}$ is called the **supremum**, or **least upper bound** if

1. m is an upper bound of S
2. $\forall m' \in \mathbb{R}$, if $m' < m$ then m' is not an upper bound of S.

Remark. $\sup[0, 1\} = 1$.

Proof. Suppose $\sup[0, 1) \neq 1$. Since 1 is an upper bound and condition 1 holds, then condition 2 must fail. Suppose $m < 1$ is an upper bound, but $\frac{1}{2}(m + 1) \in S, \frac{1}{2}(m + 1) > m$. ✓

Remarks.

1. $\sup S$ is unique.

2. $\sup S$ may not be an element of S.
3. If $\max S$ exists, then $\max S = \sup S$

Example. $S = \emptyset$, then any $x \in \mathbb{R}$ is an upper bound and lower bound of S, but $\max S, \min S, \sup S, \inf S$ do not exist.

Theorem 1.16. $\forall x, y \in \mathbb{R}, x < y$, $\exists r \in \mathbb{Q}$ s.t. $x < r < y$.

Theorem 1.17 (Completeness Axiom)**.** If $S \subset \mathbb{R}, S \neq \emptyset$, and S bounded from above, then $\sup S$ exists.

Corollary 1.18. If $S \subset \mathbb{R}, S \neq \emptyset$, and S bounded from below, then $\inf S$ exists.

Proof of CA $\iff$ corollary. Assume CA. By CA, we know the existence of $\sup S$. WTS that $\inf S$ exists, i.e. $\exists M \in \mathbb{R}$, $\forall s \in S, M \leq s$, and $\forall M' > M, \exists s \in S$ s.t. $M \leq S < M'$.

Consider $S' = \{-x, x \in S\}$. Since S nonempty, S' is nonempty. WTS S' bounded from above, i.e. $\exists M$ s.t. $\forall s \in S'$, $M \geq s$.

Since S bounded from below, $\exists M \in \mathbb{R}, \forall s \in S$, $s \geq M$. Pick arbitrary $s \in S', -s \in S \implies M \leq -s \iff -M \geq s$. Hence S' bounded from above. By CA, $\sup S'$ exists, say M_2.

We claim that $\inf S = -M_2$.

- We first show $\forall s \in S, s \geq -M_2$. Fix $s \in S$ arbitrary,

$$\begin{aligned} s \in S &\implies -s \in S' \\ &\implies -s \leq M_2 \\ &\implies s \geq -M_2 \end{aligned}$$

- We then show that any $M' > -M_2$ is not a lower bound. Take arbirary $M' > -M_2$

$$\begin{aligned} M' > -M_2 &\implies -M' < M_2 \\ &\implies \exists s \in S' \text{ s.t } -M' < s \leq M_2 \\ &\implies M' > -s \geq M_2 \end{aligned}$$

  Since $s \in S' \iff -s \in S$, M' is not a lower bound of S.

✓

Remark. The completeness axiom does not hold in $\mathbb{Q}$.

Proof. Take $S = \{x \in \mathbb{Q} : x \geq 0, x^2 < 2\}$.
We claim that any $m \in \mathbb{Q}$ that upper bounds S cannot be its supremum. **Exercise**. ✓

Remark (Implications of CA)**.**
1. $\mathbb{Q}$ dense in $\mathbb{R}$.
2. $\mathbb{R} \setminus \mathbb{Q}$ dense in $\mathbb{R}$.
3. $x^2 = 2$ has solution in $\mathbb{R}$, i.e. $\sqrt{2} \in \mathbb{R}$
4. (CCC) If a sequence of real numbers converges, its limit is in $\mathbb{R}$, this statement is equivalent to CA.

Theorem 1.19 (Archimedean Property)**.** $\mathbb{N}$ is unbounded from above in $\mathbb{R}$.

Proof. Suppose $\mathbb{N}$ is bounded from above by $m \in \mathbb{R}$.

$$\begin{aligned} &\implies \exists \sup \mathbb{N} \in \mathbb{R} \text{ by CA} \\ &\implies m' := \sup \mathbb{N} \text{ is LUB} \\ &\implies m' - 1 \text{ is not an upper bound} \\ &\implies \exists n \in \mathbb{N} \text{ s.t. } m' - 1 < n \\ &\implies m' < n + 1 \in \mathbb{N} \end{aligned}$$

This contradicts with the assumption that m' is the supremum. ✓

Proposition 1.20 (Equivalent statements to Archimedean)**.**
1. $\forall x > 0, y \in \mathbb{R}$, $\exists n \in \mathbb{N}$ s.t. $nx > y$

2. $\forall x > 0, \exists n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < x$

Proof.

1. AP $\implies$ (1): Consider $\frac{y}{x} \in \mathbb{R}$. By AP, $\exists n \in \mathbb{N}$ s.t. $n > \frac{y}{x}$. Hence $nx > y$.
2. (1) $\implies$ (2): Take $y = 1$, then $\exists n \in \mathbb{N}$ s.t. $nx > 1 \implies x > \frac{1}{n} > 0$.
3. (2) $\implies$ (AP): By contrapositive. WTS $\mathbb{N}$ bounded from above $\implies \exists x \in \mathbb{R}, x > 0, \forall n \in \mathbb{N}, \frac{1}{n} > x$.

   Since $\mathbb{N}$ bounded, let n_0 denote an upper bound. Then let $x_0 = \frac{1}{n_0} > 0$. We have for arbitrary n

$$n < n_0 \implies \frac{1}{n} > \frac{1}{n_0} = x_0$$

✓

Result 1.21 (Implication of CA)**.** $x^2 = 2, x > 0$ has a unique solution in $\mathbb{R}$, called $\sqrt{2}$

Proof. Let $S := \{x \in \mathbb{R}, x > 0, x^2 < 2\}$. We claim that

- $S \neq \emptyset$. Take $1, 1 > 0, 1^2 < 2$. Check.
- S is bounded from above. Take 2. Check. Suppose $\exists x \in S, x > 0$ then $x^2 > 2$.

Hence by CA, S has a supremum in $\mathbb{R}$, say m.

Claim: $m > 0, m^2 = 2$. It suffices to check $m^2 = 2$. ATC otherwise.

1. If $m^2 < 2$, then we can find $n \in \mathbb{N}$ s.t. $\left(m + \frac{1}{n}\right)^2 < 2$.

   By AP, $\exists n \in \mathbb{N}$ s.t. $n > \frac{2m+1}{2-m^2}$, then

$$\begin{aligned} n > \frac{2m+1}{2-m^2} &\implies \frac{2m+1}{n} < 2 - m^2 \\ &\implies m^2 + \frac{2m}{n} + \frac{1}{n} < 2 - m^2 \\ &\implies m^2 + \frac{2m}{n} + \frac{1}{n} < 2 \\ &\implies m^2 + \frac{2m}{n} + \frac{1}{n^2} < m^2 + \frac{2m}{n} + \frac{1}{n} < 2 \\ &\implies \left(m + \frac{1}{n}\right)^2 < 2 \end{aligned}$$

2. Assume $m^2 > 2$, then we can find $n \in \mathbb{N}$ such that $m - \frac{1}{n}$ is also an upper bound of S.

   It suffices to find an n such that $\left(m - \frac{1}{n}\right)^2 > 2$, which implies $\left(m - \frac{1}{n}\right)^2 > x^2 \implies m - \frac{1}{n} > x, \forall x \in S$.

   By AP, $\exists n \in \mathbb{N}$ s.t. $n > \frac{2m}{m^2-2}$, then

$$\begin{aligned} n > \frac{2m}{m^2-2} &\implies m^2 - 2 > \frac{2m}{n} \\ &\implies m^2 - \frac{2m}{n} > 2 \\ &\implies m^2 - \frac{2m}{n} + \frac{1}{n^2} > m^2 - \frac{2m}{n} > 2 \\ &\implies \left(m - \frac{1}{n}\right)^2 > 2 \end{aligned}$$

✓

Definition 1.22 (Density)**.** $A \subset B$ is dense if $\forall x \in B, \forall \epsilon > 0, \exists y \in A$ s.t. $|x - y| < \epsilon$.

Remark. We say A is dense in B if you can do ϵ-approximation for anything in B with something in A.

Theorem 1.23. We have

1. $\mathbb{Q}$ is dense in $\mathbb{R}$.
2. $\mathbb{R} \setminus \mathbb{Q}$ is dense in $\mathbb{R}$.

Remark. $\mathbb{Q}$ dense in $\mathbb{R} \iff \forall a \in \mathbb{R}, b \in \mathbb{R}, a < b, \exists r \in \mathbb{Q}$ s.t. $a < r < b$.

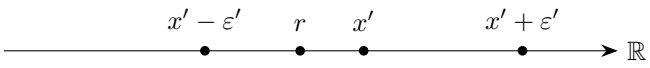
Remark. $r \in \mathbb{Q}, r \neq 0 \implies \sqrt{2}r \in \mathbb{R} \setminus \mathbb{Q}$

Proof. Thm 1.19.

1. We first show that (1) $\implies$ (2), i.e. $\mathbb{Q}$ dense $\implies$ $R \setminus \mathbb{Q}$ dense.

   $\forall x \in \mathbb{R}, \forall \epsilon > 0$. WTS $\exists y \in \mathbb{R} \setminus \mathbb{Q}$ s.t. $|x - y| < \epsilon$.

   Let $x' = \frac{x}{\sqrt{2}}$, $\epsilon' = \frac{\epsilon}{\sqrt{2}}$. Since $\mathbb{Q}$ dense in $\mathbb{R}$, $\exists r \in \mathbb{Q}$ s.t. $r \neq 0, |x' - r| < \epsilon'$. (If $r = 0$, then we can pick a smaller window $\epsilon'' = \frac{r - x'}{2}$.)

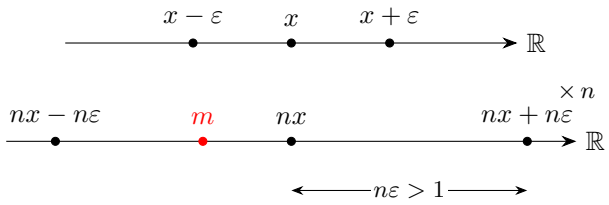
   

   Then, we take $y = \sqrt{2}r$.

   $$\begin{aligned} |x - \sqrt{2}r| &= \sqrt{2}|x' - r| \\ &< \sqrt{2}\epsilon' \\ &= \epsilon \end{aligned}$$

2. Now we show (1), i.e. $\mathbb{Q}$ is dense in $\mathbb{R}$. I.e. WTS $\forall x \in \mathbb{R}, \forall \varepsilon > 0$ $\exists y \in \mathbb{Q}$ s.t. $|x - y| < \varepsilon$.

   By Archimedean property, $\exists n \in \mathbb{N}$s.t.$n\varepsilon > 1$

   Also $\forall y \in \mathbb{R}, \exists m \in \mathbb{Z}$s.t.$|y - m| < 1$ Hence take $y = nx$

   $$\begin{aligned} & |nx - m| < 1 < n\varepsilon \\ \implies & \left| x - \frac{m}{n} \right| < \frac{n\varepsilon}{n} = \varepsilon \end{aligned}$$

✓

Side note: Cardinality

Definition 1.24 (Finite)**.** A set $S \neq \emptyset$ is **finite** if $\exists n \in \mathbb{N}$ and a bijection $f : S \to \{1, 2, \dots n\}$. Otherwise S is **infinite**.

Definition 1.25 (Countability)**.**

1. A set S is **countably infinite** if there is a bijection $f : S \to \mathbb{N}$
2. A set S is **countable** if it is empty or countably infinite
3. A set S is **uncountable** if it is not countable.

Remark. Every nonempty finite subset of $\mathbb{R}$ has a max and min.

Proposition 1.26.

1. Any subset of a countable set is countable.
2. A countable union of countable sets is countable.
3. $\mathbb{Q}$ is countable. **Exercise**
4. $\mathbb{R} \setminus \mathbb{Q}$ is uncountable $\iff$ $\mathbb{R}$ is uncountable. **Exercise**
5. Union of a countable set and an uncountable set is uncountable.

Proof that $\mathbb{Q}$ is countable. There is an injection from $\mathbb{Q}$ to the set $A \coloneqq \{(a, b) : a, b \in \mathbb{Z}\}$. Fix arbitrary $a \in \mathbb{Z}$, $A_a \coloneqq \{(a, b) : b \in \mathbb{Z}\}$. There is a bijection between A_a and $\mathbb{Z}$, and $\mathbb{Z}$ countable, hence A_a is countable. $A = \bigcup_{a \in \mathbb{Z}} A_a$ is a countable union of countable sets and is therefore countable. ✓

2 Metric spaces and point-set topology

2.1 Metric spaces, open and closed sets

Definition 2.1 (Metric space)**.** A **metric space** (X, d) is a set X of elements (called points) equipped with a metric or function $d : X \times X \to [0, \infty)$ such that

1. $\forall x, y \in X, d(x, y) \geq 0$. i.e. The image of d is in $\mathbb{R}$ and non-negative
2. $d(x, y) = 0 \iff x = y$
3. $\forall x, y \in X, d(x, y) = d(y, x)$
4. $\forall x, y, z, \in X,\ d(x, z) \leq d(x, y) + d(y, z)$

Definition 2.2 (Discrete metric)**.** The **discrete metric** d is defined such that $\forall x, y$

$$d(x, y) = \mathbf{1}[x \neq y] = \begin{cases} 0, & \text{if } x = y \\ 1, & \text{if } x \neq y \end{cases}$$

Example. The following are metric spaces

- $X = \mathbb{R}, d(x, y) = |x - y|$
- $X = \mathbb{R}^n := \{(x_1, x_2, \dots x_n) : x_i \in \mathbb{R}\}$, $d_1(x, y) = \left(\sum_{i=1}^{n} |x_i - y_i|^2 \right)^{\frac{1}{2}}$, i.e. Euclidean distance
- $X = \mathbb{R}^n$, $d_2(x, y) = \max_i |x_i - y_i|$
- X is the set of continuous functions on $[a, b]$, and $d(f, g) = \sup_{x \in [a,b]} |f(x) - g(x)|$
- Any $X \neq \emptyset$ with the discrete metric.

Definition 2.3 (Neighborhood)**.** Let (X, d) be a metric space. Let $x \in X, \varepsilon > 0$. The **ε-neighborhood** of x is

$$N(x, \varepsilon) = B_\varepsilon(x) := \{y \in X, d(x, y) < \varepsilon\}$$

A **deleted neighborhood** is

$$\begin{aligned} N^*(x, \varepsilon) :&= \{y \in X, 0 < d(x, y) < \varepsilon\} \\ &= N(x, \varepsilon) \setminus \{x\} \end{aligned}$$

Definition 2.4 (Interior)**.** Let (X, d) be a metric space, $S \subset X$. A point $x \in S$ is an **interior point** of S if

$$\exists \varepsilon > 0 \text{ s.t. } N(x, \varepsilon) \subset S$$

I.e. If an ε-neighborhood around x is in S, then x is an interior point.

The set of all interior points of S is called the **interior** of S, denoted $\operatorname{int} S$

$$\operatorname{int} S := \{x \in S : \exists \varepsilon > 0 \text{ s.t. } N(x, \varepsilon) \subset S\}$$

Definition 2.5 (Boundary)**.** Let (X, d) be a metric space, $S \subset X$. A point $x \in S$ is a **boundary point** of S if $\forall \varepsilon > 0$,

$$\begin{aligned} N(x, \varepsilon) \cap S \neq \emptyset \\ N(x, \varepsilon) \cap S^c \neq \emptyset \end{aligned}$$

The set of boundary pints of S is $\operatorname{bd} S$

$$\begin{aligned} \operatorname{bd} S :&= \{x \in X : \forall \varepsilon > 0, N(x, \varepsilon) \cap S \neq \emptyset, N(x, \varepsilon) \cap S^c \neq \emptyset\} \\ &= (\operatorname{int} S \cup \operatorname{int} S^c)^c \end{aligned}$$

Example. $\operatorname{int} \mathbb{N} = \emptyset$,

Proof. Fix $x \in \mathbb{R}$ and $\varepsilon > 0$ arbitrary. Take $y = x + \frac{1}{2} \min(\varepsilon, 1)$. Then

- $|y - x| = \frac{1}{2} \min(\varepsilon, 1) \implies y \in N(x, \varepsilon)$
- $x < y < x + 1 \implies y \notin \mathbb{N}$

Hence $N(x,\varepsilon) \not\subset \mathbb{N}$ for all $\varepsilon > 0 \implies x \notin \operatorname{Int}\mathbb{N}$. ✓

Example. $\operatorname{bd}\mathbb{N} = \mathbb{N}$.

Proof.

1. $\mathbb{N} \subset \operatorname{bd} N$. Fix $x \in \mathbb{N}, \varepsilon > 0$,
   - $x \in N(x,\varepsilon) \cap \mathbb{N} \neq \emptyset$.
   - $y = x + \frac{1}{2}\min(\varepsilon, 1) \in N(x,\varepsilon) \cap \mathbb{N}^c \neq \emptyset$.
2. $\mathbb{N} \supset \operatorname{bd} N$. Fix arbitrary $x \in \operatorname{bd}\mathbb{N}$. ATC $x \notin \mathbb{N}$, then we consider 2 cases
   - if $x < 1$, then take $\varepsilon = 1 - x$. $\forall y \in N(x,\varepsilon), 1 > y \notin \mathbb{N}$, hence $N(x,\varepsilon) \cap \mathbb{N} = \emptyset$, which contradicts the definition of boundary point.
   - $if x > 1$, let $n \in \mathbb{N}$ such that $n < x < n+1$. Let $\varepsilon = \frac{1}{2}\min(x-n, n+1-x)$, then $\forall y \in N(x,\varepsilon)$, $y \notin \mathbb{N}$, hence $N(x,\varepsilon) \cap \mathbb{N} = \emptyset$, which contradicts the definition of boundary point.

✓

Remark.

1. $\operatorname{int} S = S \setminus \operatorname{bd} S$
2. boundary points may or may not be in S.
3. Any point in a set S is either a boundary or an interior point, but not both.

Definition 2.6 (Closure)**.** The **closure** of S, denoted $\overline{S}$, is defined as

$$\overline{S} := S \cup \operatorname{bd} S$$

Definition 2.7 (Openess and closedness)**.**

1. $S \subset X$ is called **open** if $S = \operatorname{int} S$
2. $S \subset X$ is called **closed** if S^c is open.

Remark. A set can be both open and closed. For example

- $\mathbb{R}$ is both open and closed.
- $\emptyset$ is both open and closed.
- $\mathbb{N}$ is not open, but closed.

Example. $S = (1,3)$ is open and not closed.

Example. Let (X,σ) be a metric space where σ is the discrete metric. Take $x_0 \in X$, $S = \{x_0\}$ is open.

To see why, let $\varepsilon = \frac{1}{2}$.

$$\begin{aligned} & N\left(x_0, \frac{1}{2}\right) = \{x_0\} \subset S \\ \implies & x_0 \in \operatorname{int} S \\ \implies & \operatorname{int} S = S \\ \implies & S \text{ open} \end{aligned}$$

Hence any point in a set is interior, and any set is open and closed.

Proposition 2.8.

1. S open $\iff \operatorname{bd} S \subset S^c$
2. S closed $\iff \operatorname{bd} S \subset S \iff S = \overline{S} := S \cup \operatorname{bd} S$
3. $\overline{S}$ is closed
4. $\operatorname{int} S$ is open $\implies \operatorname{int}(\operatorname{int}(S)) = \operatorname{int}(S)$

Proof. S **closed** $\iff \operatorname{bd} S \subset S$**.**

By definition, S closed $\iff$ S^c open. Also by definition, $\operatorname{bd} S = \operatorname{bd} S^c$.

$$\begin{aligned}
\operatorname{bd} S \subset S &\iff \operatorname{bd} S^c \subset S && \text{by definition} \\
&\iff \operatorname{bd} S^c \cap S^c = \emptyset \\
&\iff \forall x \in S^c, x \notin \operatorname{bd} S^c \\
&\iff \forall x \in S^c, x \in \operatorname{int} S^c \\
&\iff S^c = \operatorname{int} S^c \\
&\iff S^c \text{ open} \\
&\iff S \text{ closed}
\end{aligned}$$

✓

Proposition 2.9.

1. The union of open sets is open.
2. The intersection of finitely many open sets is open.
3. The intersection of closed sets is closed.
4. The union of finitely many closed sets is closed.

Example. $S_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$ is open for all $n \in \mathbb{N}$. $\bigcap_{n=1}^{\infty} S_n = \{0\}$ is closed.

To see why the intersection is closed, $\forall x > 0$, by AP, $\exists n \in \mathbb{N}$ s.t. $0 < \frac{1}{n} < x$. Hence $x \notin S_n$ and $S \notin \bigcap S_n$.

At the same time, $0 \in S_n, \forall n \in \mathbb{N}$.

Since $0 \notin \operatorname{int}\{0\}$, $\{0\}$ is not open.

Proof of theorem 2.9.1 $\iff$ 2.9.3. Fix arbitrary open sets $S_\alpha, \alpha \in \Lambda$.

$$\bigcup_\alpha S_\alpha \text{ open} \iff \left(\bigcup_\alpha S_\alpha\right)^c \text{ closed} \iff \bigcap_\alpha S_\alpha^c \text{ closed}$$

✓

Proof of theorem 2.9.2. WTS intersection of finitely many open sets is open.

Let $S_1, \dots S_n$ be open sets, $S = \bigcap_{k=1}^n S_k$. WTS $\forall x S, \exists \varepsilon \in \mathbb{R}, \varepsilon > 0$ s.t. $N(x, \varepsilon) \subset S$. Note that

$$N(x, \varepsilon) \subset S \iff N(x, \varepsilon) \subset S_i, \forall i = 1, \dots, n$$

Since $x \in S_i$ and S_i is open, $\exists \varepsilon_i > 0$ s.t. $N(x, \varepsilon_i) \subset S_i$. Let $\varepsilon = \min\{\varepsilon_1, \dots, \varepsilon_n\}$, and we know such ε exists because a sequence of finitely many numbers has a minimum. ✓

Definition 2.10 (Accumulation points and isolated points)**.**

1. Let $S \subset X$, a point $x \in X$ is an **accumulation point** or limit point of S if $\forall \varepsilon > 0$,

$$N^*(x, \varepsilon) \cap S \neq \emptyset$$

   The set of all accumulation points of S is denoted S'.
2. If $x \in S \setminus S'$, then x is an **isolated point** of S.

Remark. $x \in X$ is an accumulation point of $S \subset X$ if and only if every neighbrhood of x contains infintely many points in S.

Example. $S = \{0\} \cup (1, 2)$, $S' = [1, 2]$

Example. $S = \mathbb{Q} \cap (1, 2)$, $S' = [1, 2]$

Proposition 2.11.

1. $\overline{S} = S \cup S'$
2. S close $\iff$ $S' \subset S$
3. $\overline{S}$ is the smallest closed set that contains S.

Proof of 2.11.

1. WTS $\overline{S} = S \cup S' \iff S \cup \operatorname{bd} S = S \cup S'$
   (a) We first show that $LHS \subset RHS$. $\forall x \in S \cup \operatorname{bd} S$, $x \in S \cup S'$.

- If $x \in S$, done.
- If $x \in \operatorname{bd} S \cap S^c$, WTS $x \in S'$, i.e. $\forall \varepsilon > 0, N^*(x,\varepsilon) \neq \emptyset$.

  Since $x \in \operatorname{bd} S$, we know $N(x,\varepsilon) \cap S \neq \emptyset$. Since $x \in S^c$, we know $\exists y \in N(x,\varepsilon) \cap S, y \neq x$. Hence $y \in N^*(x,\varepsilon) \cap S$, and $N^*(x,\varepsilon) \neq \emptyset$.

(b) We then show $RHS \subset LHS$, i.e. $\forall x \in S \cup S', x \in S \cup \operatorname{bd} S$.

- If $x \in S$, done.
- If $x \in S' \setminus S$, WTS $x \in \operatorname{bd} S$. $\forall \varepsilon > 0, x \in N(x,\varepsilon) \cap S^c \implies N(x,\varepsilon) \cap S^c \neq \emptyset$. $x \in S', N^*(x,\varepsilon) \cap S \neq \emptyset \implies N(x,\varepsilon) \cap S \neq \emptyset$.

2. WTS S close $\iff S' \subset S$. Recall

$$S \text{ closed} \iff S = \operatorname{bd} S \subset S \iff S = \bar{S}$$

- S closed $\implies S = \bar{S} = S \cup S' \implies S' \subset S$
- $S' \subset S \implies \bar{S} = S \cup S' = S \implies S$ closed.

✓

Remark. 1. $S' = (S \cup \operatorname{bd} S) \setminus \{\text{isolated points}\}$

2. Isolated points may not be boundary points.
   - In discrete metric space, any $S \subset X$, $s \in S$ is isolated and interior.

2.2 Compact sets

Definition 2.12 (Compactness)**.** A set $S \in X$ is **compact** if for any open cover of S has a finite subcover.

An open cover F is a set of open sets $O_\alpha, \alpha \in \Lambda$ s.t. $\bigcup_{\alpha\in\Lambda}^n O_\alpha \supset S$, and a subcover is a finite subcollection $O_{\alpha_1}, \dots O_{\alpha_n} \in F$ s.t. $\bigcup_{i=1}^n O_{\alpha_i} \supset S$.

If X itself is a compact set, then we say (X,d) is a compact metric space.

Example. $S = [1,\infty)$ is not compact.
We can show that there exists an open cover F such that no finite subcollection of it covers S.
Let $O_n = (0,n), \forall n \in \mathbb{N}$. We observe that

$$S = [1,\infty) \subset \bigcup_{n=1}^{\infty} O_n$$

We claim that for any $O_{n_1}, \dots O_{n_k}$,

$$S \not\subset \bigcup_{i=1}^{k} O_{n_i}$$

Let $n_0 = \max\{n_1, \dots n_k\}$, then

$$\bigcup_{i=1}^{n} O_{n_i} \subset (0, n_0)$$

Hence $\exists x \in S$ s.t. $x > n_0$ and $x \notin \bigcup_{i=1}^k O_{n_i}$.

Proposition 2.13. A closed subset of a compact set is compact.

Proof. Let S be compact, and let $H \subset S$ be a closed subset. WTS H is compact.

Let $\{O_\alpha\}_{\alpha\in\Lambda}$ be any open cover of H. Since H is closed, we know H^c open. So

$$\{O_\alpha\}_{\alpha\in\Lambda} \cup \{H^c\}$$

is an open cover of S.

Since S is compact, there exists N and $O_{\alpha_1}, \dots O_{\alpha_N}$ s.t.

$$\begin{aligned} & O_{\alpha_1} \cup \dots O_{\alpha_N} \cup H^c \supset S \supset H \\ \implies & O_{\alpha_1} \cup \dots O_{\alpha_N} \supset H \end{aligned}$$

and hence we have found a finite subcover of H. Hence H is compact. ✓

Definition 2.14 (Boundedness for sets)**.** We say that a set $S \subset X$ is **bounded** if $\exists M > 0$ s.t. $d(x,y) \leq M, \forall x, y \in S$.

Remark. A set is bounded is some ε ball around some point contains all points.

Theorem 2.15 (Heine-Borel)**.** A set $S \subset \mathbb{R}$ is compact if and only if it is bounded and closed.

Remarks. 1. The forward direction is true in any metric space. The converse is true for $\mathbb{R}$ but not necessarily all metric spaces.
2. The converse is equivalent to CA.

Proof.

1. "Compact $\implies$ bounded and closed for any metric space"
   (a) "Compact $\implies$ bdd"

   Fix $x_0 \in S$, then $\{N(x_0, r)\}_{r \geq 0}$ is an open cover of S.

   Since S cpt, $\exists r_1, \dots r_N$ s.t.
   $$N(x_0, r_1) \cup N(x_0, r_N) \supset S$$
   Let $M = 2\max\{r_1, \dots r_N\}$, $M > 0$. Then
   $$S \subset N\left(x_0, \frac{M}{2}\right)$$

   (b) "Compact $\implies$ closed". Let $S \subset X$ be cpt. WTS that $S' \subset S$. Assume $S' \not\subset S \implies x_0 \in S' \setminus S$. Let $O_\varepsilon = \{x \in X : d(x, x_0) > \varepsilon\}, \varepsilon > 0$. Then
   $$\bigcup_{\varepsilon > 0} O_\varepsilon \supset S$$
   Since S cpt, $\exists \varepsilon_1, \dots \varepsilon_N > 0$ s.t. $O_{\varepsilon_1} \cup \dots \cup O_{\varepsilon_N} \supset S$. Let $\varepsilon_{\min} = \frac{1}{2}\min\{\varepsilon_1, \dots \varepsilon_N\}$. Then
   $$N(x_0, \varepsilon_{\min}) \cap S = \emptyset$$
   Which contradicts $x_0 \in S'$.

2. "Bounded and closed $\implies$ compact in $\mathbb{R}$"

   Let $\mathcal{F} = \{O_\alpha\}_{\alpha \in \Lambda}$ be any open cover of S. $\forall x \in \mathbb{R}$, define $S_x := S \cap (-\infty, x]$, define B
   $$B \coloneqq \{x \in \mathbb{R} : S_x \text{ covered by finitely many } O_\alpha\text{'s in } \mathcal{F}\}$$
   We first observe that $B \neq \emptyset$. Since S bounded,
   $$\implies \exists m \in \mathbb{R} \text{ s.t. } m \text{ lower bounds } S$$
   $$\implies \forall x < m, S_x = S \cap (-\infty, x] = \emptyset$$
   It suffices to show that B is unbounded from above. If this is true, since S bounded from above, let M be an upper bound of S. Then $\exists b \in B$ s.t. $b > M$ and therefore $S_b \cap (-\infty, b] = S$, and S has a finite subcover.

   WTS that B is unbounded from above. ATC otherwise, then by CA, $\sup B$ exists, say b.
   (a) Case 1: $b \in S$

   $\exists O_{\alpha_0} \in \mathcal{F}$ s.t. $b \in O_{\alpha_0}$.

   Since O_{α_0} open, $\exists \varepsilon > 0$ s.t. $N(b, \varepsilon) \subset O_{\alpha_0}$.

   By definition of supremum, $\exists b' \in B$ s.t. $b - \varepsilon < b' < b$.

   By definition of B, $\exists \alpha_1, \dots \alpha_N$ s.t.
   $$\begin{aligned} & O_{\alpha_1} \cup \dots O_{\alpha_N} \supset S_{b'} = S \cap (\infty, b'] \\ \implies & O_{\alpha_1} \cup \dots O_{\alpha_N} \cup O_{\alpha_0} \supset S_{b+\varepsilon} = S \cap (\infty, b + \varepsilon] \\ \implies & b + \varepsilon \in B \\ \implies & b \text{ is not upper bound of} B \end{aligned}$$

   (b) Case 2: $b \notin S$, $b \in S^c$.

   Since S closed, S^c open, therefore $\exists \varepsilon > 0$ s.t. $N(b, \varepsilon) \subset S^c$. By definition of supremum, $\exists b' \in B$ s.t. $b - \varepsilon < b' < b$. Hence $S^{b'}$ has finite open cover.

   Since $S_{b+\varepsilon/2} = S_{b'}$, $S_{b+\varepsilon/2}$ also has finite open cover, i.e. $b + \varepsilon/2 \in B$, contradicting the assumption that b is an upper bound.

   ✓

Theorem 2.16 (Bolzano-Weiestrass)**.** If $S \subset \mathbb{R}$ is bounded and contains infinitely many points then $S' \neq \emptyset$.

Remark. Bolzano-Weierstrass gives the existence of limit points, but it does not tell us that the limit points are in S.

Proof. Suppose $S' = \emptyset$, then $S = \bar{S}$ is closed. By HB, S is compact.

For any $x \in S, x \notin S'$, x must be isolated point, i.e.

$$\exists \varepsilon_x > 0, s.t. N(x, \varepsilon_x) \cap S = \{x\}$$

Then $\{N(x, \varepsilon_x)\}_{x \in S}$ is an open cover of S. By compactness, $\exists x_1, x_2, \dots x_N$ s.t.

$$N(x_1, \varepsilon_{x_1}) \cup \dots N(x_N, \varepsilon_{x_N}) \supset S$$

However, each one of the neighborhoods contains only its center, i.e.

$$N(x_1, \varepsilon_{x_1}) = \{x_1\}, \dots N(x_n, \varepsilon_{x_N}) = \{x_N\}$$

Hence

$$\{x_1\} \cup \dots \{x_N\} \supset S$$

and this contradicts S having infinitely many points. ✓

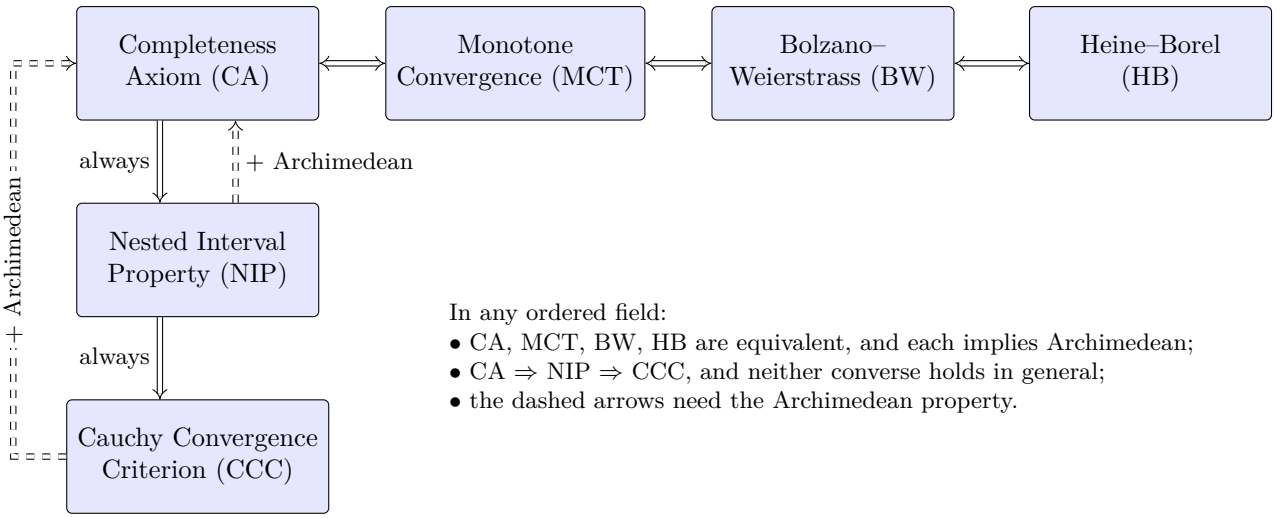
Remark. S has infinite points and $S' \neq \emptyset \not\Longleftrightarrow S$ bounded.

Proof. Let $S \coloneqq [1, 2] \cup \mathbb{N}$. Then $S' = [1, 2] \neq \emptyset$, but S unbounded. ✓

Remark. If $S \subset \mathbb{R}$, and every infinite subset of $\mathbb{S}$ has an accumulation point in $\mathbb{R}$, then S is bounded.

Proof. **To be added in** ✓

Remark (Relationship between results)**.**



In any ordered field:
- CA, MCT, BW, HB are equivalent, and each implies Archimedean;
- CA ⇒ NIP ⇒ CCC, and neither converse holds in general;
- the dashed arrows need the Archimedean property.

Theorem 2.17 (Nested Interval Property)**.** Let $\{I_n = [a_n, b_n] : n \in \mathbb{N}\}$ be a family of closed bounded intervals s.t. $I_{n+1} \subset I_n$ and $I_n \neq \emptyset$, $\forall n \in \mathbb{N}$. Then

$$\bigcap_{i=1}^{\infty} I_n \neq \emptyset$$

Theorem 2.18 (Generalized NIP)**.** If $\mathcal{F} = \{K_\alpha\}$ is a collection of compact subsets of X such that the intersection of every finite subcollection of $\mathcal{F}$ is nonempty, then

$$\bigcap_{\alpha} K_\alpha \neq \emptyset$$

Remark (Theorem 2.18 $\implies$ theorem 2.17)**.** Let $\mathcal{F} = \{I_n : n \in \mathbb{N}\}$ be as in Theorem 2.17, with $X = \mathbb{R}$. We check the hypotheses of Theorem 2.18.

Compactness. Each $I_n = [a_n, b_n]$ is closed and bounded in $\mathbb{R}$, hence compact by Heine-Borel.

Finite intersections are nonempty. Take any finite subcollection $I_{n_1}, \dots, I_{n_k}$ and let $N = \max\{n_1, \dots, n_k\}$. Since

the intervals are nested, $I_m \subset I_n$ whenever $m \geq n$, so $I_N \subset I_{n_j}$ for every j. Therefore

$$I_{n_1} \cap \ldots \cap I_{n_k} = I_N \neq \emptyset$$

by assumption.

Hence Theorem 2.18 applies and gives

$$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$$

Proof of 2.18. Suppose $\bigcap_\alpha K_\alpha = \emptyset$. Fix any $K_{\alpha_0} \in \mathcal{F}$. K_{α_0} is compact by assumption. Fix $x \in K_{\alpha_0}$.

$$x \notin \bigcap_\alpha K_\alpha \implies \exists \alpha_x \text{ s.t. } x \notin K_{\alpha_x} \iff x \in K_{\alpha_x}^c$$

Also K_{α_x} compact $\implies$ K_{α_x} closed $\implies$ $K_{\alpha_x}^c$ open. Hence $\{K_{\alpha_x}\}_{x \in K_{\alpha_0}}$ is an open cover of K_{α_0}. By compactness of K_{α_0}, $\exists \alpha_1, \ldots \alpha_N$ s.t.

$$\begin{aligned} & K_{\alpha_0} \subset K_{\alpha_1}^c \cup \ldots K_{\alpha_N}^c \\ \iff & K_{\alpha_0} \cap (K_{\alpha_1}^c \cup \ldots K_{\alpha_N}^c)^c = \emptyset \\ \iff & K_{\alpha_0} \cap K_{\alpha_1} \cap \ldots K_{\alpha_N} = \emptyset \end{aligned}$$

which contradicts the assumption that any finite subcollection has a nonempty intersection. ✓

2.3 Connected sets

Definition 2.19 (Disconectedness / connectedness)**.** A set $S \subset X$ is **disconnected** if $\exists A, B$ s.t.

- A, B open
- $S \subset A \cup B$
- $A \cap S \neq \emptyset$
- $B \cap S \neq \emptyset$
- $A \cap B = \emptyset$

Example.

1. $(1,2) \cup (2,3)$ disconnected, take $A = (0,2), B = (2,\infty)$
2. $\mathbb{Q}$ disconnected, take $A = (-\infty, \sqrt{2}), B = (\sqrt{2}, \infty)$
3. For a discrete metric space, connected sets are $\emptyset$ or singleton, any other set is disconnected.
4. Any singleton set is connected.

Definition 2.20. A set $S \subset \mathbb{R}$ is an **interval** if $\forall a, b \in \mathbb{R}, a < b$

$$a \in S, b \in S \implies [a,b] \subset S$$

Theorem 2.21 (Connected sets in $\mathbb{R}$)**.** $S \subset \mathbb{R}$ is connected if and only if one of the following holds

1. $S = \emptyset$
2. S singleton
3. S is an interval.

Proof.

1. $\implies$: Assume $S \subset \mathbb{R}$ connected and contains at least 2 points. WLOG, $a < b, a \in S, b \in S$. WLOG, assume $\inf S, \sup S \notin S$. WTS $S = (\inf S, \sup S)$.
   - $S \subset (\inf S, \sup S)$ is for free.
   - Suffices to show $(\inf S, \sup S) \subset S$

     Suppose $\exists x \in (\inf S, \sup S) \cap S^c$, then let $A := (\infty, x), B = (x, \infty)$. Then
     - $S \subset A \cup B$
     - $A \cap B = \emptyset$
     - A, B open
     - $A \cap S \neq \emptyset$ because by definition of infimum, $\exists y \in S$ s.t. $\inf S < y < x$
     - $B \cap S \neq \emptyset$ because $\exists y \in S$ s.t. $\sup S > y > x$.
2. $\impliedby$

✓

Definition 2.22 (Separatedness)**.** Two sets $A, B \subset X$ are **separated** if

$$\bar{A} \cap B = \emptyset, A \cap \bar{B} = \emptyset$$

Remark. A, B separated may not necessarily have $\bar{A} \cap \bar{B} = \emptyset$, e.g.

$$A = (0, 1), B = (1, 2)$$

Remark.
- Separate $\implies$ Disjoint
- Separate $\not\impliedby$ Disjoint, e.g.

$$A = (0, 1], B = (1, 2)$$

Definition 2.23 (Relatively open / closed sets)**.** Let $S \subset X$, a set $E \subset S$ is **relatively open** in S if there is an open set $U \subset X$ s.t. $E = S \cap U$.

E is **relatively closed** in S if there is a closed set $U \subset X$ s.t. $E = S \cap U$.

Theorem 2.24. Let $S \subset X$, TFAE

1. S is disconnected
2. $\exists$ nonempty sets $A, B \subset S$ s.t. A, B relatively open in S, $A \cap B = \emptyset, S = A \cup B$.
3. $\exists$ nonempty sets $A, B \subset S$ s.t. A, B separated, and $S = A \cup B$

3 Numerical Sequences

Definition 3.1 (Sequence)**.** A **sequence** $\{a_n : n \in \mathbb{N}\} = \{a_1, a_2, \dots a_n, \dots\}$ is a function $f : \mathbb{N} \to \mathbb{R}$.

3.1 Convergent sequences

Definition 3.2 (Convergence)**.** A sequence $\{a_n\}$ **converges** to $L \in \mathbb{R}$ if $\forall \varepsilon > 0$, $\exists N > 0$ s.t. $\forall n > \mathbb{N}$

$$|a_n - L| < \varepsilon$$

Then we say that L is the **limit** of $\{a_n\}$, and denote

$$\lim_{n\to\infty} a_n = L$$

If $\{a_n\}$ doesn't converge to any real number, we say $\{a_n\}$ **diverges**.

Example. $\{(-1)^n\}$ diverges.

Proof. Suppose $\lim_{n\to\infty} a_n = L$. Let $\varepsilon = \frac{1}{100}$. Then either 1 or -1 is not in $N(L, \varepsilon)$. ✓

Theorem 3.3 (Uniqueness of limit)**.** If a sequence converges then the limit is unique.

Proof. ATC otherwise, then $\{a_n\}$ converges to two different limits, $L_1 \neq L - 2$. Let $\varepsilon = \frac{1}{3}|L_1 - L_2| > 0>$ By definition of limit,

- $\exists N_1 > 0$ s.t. $\forall n > N_1, |a_n - L_1| < \varepsilon$
- $\exists N_2 > 0$ s.t. $\forall n > N_2, |a_n - L_2| < \varepsilon$

Then let $n > \max\{N_1, N_2\}$, then $|a_n - L_1| < \varepsilon, |a_n - L_2| < \varepsilon$, and

$$\begin{aligned} |L_1 - L_2| &= |L_1 - a_n + a_n - L_2| \\ &\leq |L_1 - a_n| + |a_n - L_2| \\ &< 2\varepsilon \\ &= \frac{2}{3}|L_1 - L_2| \end{aligned}$$

✓

Theorem 3.4 (Convergence implies bounded)**.** If a sequence converges then it is bounded.

Proof. *The idea is that if a sequence converges to L, past a certain point, everything is close to L. We only have to deal with finitely many terms before the sequence gets close to L, in which case we can take a max.*
Assume $\lim_{n\to\infty} a_n = L$, and let $\varepsilon = 1$. Then $\exists N \in \mathbb{N}$ s.t. $|a_n - L| < 1, \forall n > N$. Hence, $\forall n \geq N$

$$\begin{aligned} |a_n| &= |a_n - L + L| \\ &\leq |a_n - L| + |L| \\ &< 1 + |L| \end{aligned}$$

Let $M := \max\{|a_1|, |a_2|, \dots |a_N|, |L| + 1\}$. Then $|a_n| \leq M, \forall n > N$. ✓

Proposition 3.5 (Limits and accumulation points)**.**

1. If $\lim_{n\to\infty} a_n = L$ and if $\{a_n\}$ attains infinitely many values, then L is an accumulation point of $\{a_n\}$.
2. If $L \in S'$ then S contains a sequence $\{a_n\}$ that converges to L.

Proof.

1. $a_n \to L \implies N(L, \varepsilon)$ *contains all but finitely many terms.* Fix $\varepsilon > 0$. WTS $\exists x \in \{a_n\} \cap N^*(L, \varepsilon)$ Since $\lim_{n\to\infty} a_n = L, \exists N > 0$ s.t. $a_n \in N(L, \varepsilon), \forall n > N$.

   ATC otherwise, then $a_n = L, \forall n > N$. Then $\{a_n\}$ contains only $a_1, a_2, \dots, a_N, L$, which contradicts the assumption that $\{a_n\}$ has infinitely many values.

2. Let $\varepsilon_1 = 1$. Since L is an accumulation point,

$$L \in S' = \Longrightarrow \exists a_1 \in S \cap N^*(L, \varepsilon_1)$$

Repeat this, setting $\varepsilon_n = \frac{1}{n}$, and picking $a_n \in S \cap N^*(L, \varepsilon_n)$.

We claim that $\lim a_n = L$. Fix any $\varepsilon > 0$, by AP, $\exists N \in \mathbb{N}$ s.t. $\frac{1}{N} < \varepsilon$, then $\forall n > N$,

$$|a_n - L| < \frac{1}{n} < \frac{1}{N} < \varepsilon$$

✓

Remark. If $\{a_n\}$ has infinitely many values, $a_n \to L$, then L is **unique** the accumulation point.

Theorem 3.6 (Operation on limits)**.** Suppose $\lim_{n\to\infty} a_n = A$, $\lim_{n\to\infty} b_n = B$, then

1. $\lim_{n\to\infty}(a_n + b_n) = A + B$
2. $\lim_{n\to\infty} ka_n = kA$
3. $\lim_{n\to\infty} a_n b_n = AB$
4. If $b_n \neq 0, \forall n$ and $B \neq 0$, then

$$\lim_{n\to\infty} \frac{a_n}{b_n} = \frac{A}{B}$$

Proof of 3. *The idea is to use boundedness of one of the sequences to find a bound on the deviation for each of the convergent sequences from their limits, and then take the bigger N.*

Fix $\varepsilon > 0$, seek $N > 0$ s.t. $\forall n > N, |a_n b_n - AB| < \varepsilon$.

$$\begin{aligned} |a_n b_n - AB| &= |a_b b_n - a_b B + a_b B - AB| \\ &\leq |a_n b_n - a_n B| + |a_n B - AB| \\ &= |a_n|\,|b_b - B| + |a_n - A||B| \end{aligned}$$

Since convergent implies boundedness, $\exists M > 0$ s.t. $|a_n| \leq M, \forall n$. Choose M such that $|B| \leq M$.

By convergence of a_n, b_n, we can choose

- $N_1 > 0$ s.t. $|a_n - A| < \frac{\varepsilon}{2M}, \forall n > N_1$
- $N_2 > 0$ s.t. $|a_n - A| < \frac{\varepsilon}{2M}, \forall n > N_2$
- $N = \max\{N_1, N_2\}$

Then $\forall n > N$

$$\begin{aligned} |a_n b_n - AB| &\leq |a_n|\,|b_b - B| + |a_n - A||B| \\ &< |a_n| \cdot \frac{\varepsilon}{2M} + \frac{\varepsilon}{2M} \cdot |B| \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} \\ &= \varepsilon \end{aligned}$$

✓

Example. $\lim_{n\to\infty} \frac{n+1}{2n-1} = \frac{1}{2}$

By definition, fix $\varepsilon > 0$, seek $N > 0$ s.t. $\forall n > N$,

$$\left|\frac{n+1}{2n-1} - \frac{1}{2}\right| < \varepsilon$$

Simplifying the expression, we get

$$\begin{aligned} &\left|\frac{n+1}{2n-1} - \frac{1}{2}\right| = \left|\frac{2n+2-2n+1}{2(2-n-1)}\right| = \frac{3}{2}\frac{1}{2n-1} < \varepsilon \\ \iff &\frac{1}{2n-1} < \frac{2}{3}\varepsilon \\ \iff &2n-1 > \frac{3}{2}\varepsilon \\ \iff &n > \frac{1}{2}\left(\frac{3}{2}\varepsilon + 1\right) \end{aligned}$$

Take

$$N = \lceil \frac{1}{2}\left(\frac{3}{2}\varepsilon + 1\right) \rceil$$

and we are done.

Alternatively, *we can prove using algebraic properties.*

$$\begin{aligned}\frac{n+1}{2n-1} &= \frac{\frac{1}{2}(2n-1)+\frac{3}{2}}{2n-1} \\ &= \frac{1}{2} + \frac{3}{2}\frac{1}{2n-1}\end{aligned}$$

It suffices to prove $\lim_{n\to\infty} \frac{1}{2n-1} = 0$. Then by theorem, $\lim_{n\to\infty} \frac{n+1}{2n-1} = \frac{1}{2} + \frac{3}{2}\cdot 0 = \frac{1}{2}$.

Corollary 3.7. If $\lim_{n\to\infty} a_n$ converges, and $\lim_{n\to\infty} b_n$ diverges, then $\lim_{n\to\infty} a_n + b_n$ diverges.

Example. Show $\lim_{n\to\infty} a^{\frac{1}{n}} = 1$

Proof We consider 3 cases

- $a < 1$: then by theorem,

$$\begin{aligned}\lim_{n\to\infty} \frac{1}{a^{\frac{1}{n}}} &= \frac{1}{\lim_{n\to\infty} a^{\frac{1}{n}}} \\ \implies \lim_{n\to\infty} a^{\frac{1}{n}} &= \frac{1}{\lim_{n\to\infty} \frac{1}{a^{\frac{1}{n}}}} = \frac{1}{\lim_{n\to\infty} \left(\frac{1}{a}\right)^{\frac{1}{n}}}\end{aligned}$$

  and then we are in the case of $a > 1$
- $a = 1$: trivial
- $a > 1$

Hence it suffices to show for $a > 1$. Fix $\varepsilon > 0$, we seek N s.t. $\forall n > N$,

$$\begin{aligned}&\left|a^{\frac{1}{n}} - 1\right| < \varepsilon \\ \iff & a^{\frac{1}{n}} - 1 < \varepsilon \\ \iff & a^{\frac{1}{n}} < 1 + \varepsilon \\ \iff & \frac{1}{n}\ln a < \ln(1+\varepsilon) \\ \iff & n > \frac{\ln a}{\ln(1+\varepsilon)}\end{aligned}$$

Hence we can take

$$N = \lceil \frac{\ln a}{\ln(1+\varepsilon)} \rceil$$

and we are done.

Theorem 3.8 (Sandwich Theorem)**.** Let N_0 such that $a_n \le c_n \le b_n$, $\forall n > N_0$, suppose

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} b_n = L$$

then

$$\lim_{n\to\infty} c_n = L$$

Proof. Fix $\varepsilon > 0$, we seek $N > 0$ s.t. $\forall n > N$,

$$|c_n - L| < \varepsilon$$

Since $\lim a_n = L, \exists N_1$ s.t. $\forall n > N_1$, $|a_n - L| < \varepsilon$ Since $\lim b_n = L, \exists N_2$ s.t. $\forall n > N_2$, $|b_n - L| < \varepsilon$ Pick $N = \max\{N_1, N_2, N_0\}$, then $\forall n > N$,

$$\begin{aligned}c_n - L &\le b_n - L < \varepsilon \\ c_n - L &\ge a_n - L > -\varepsilon\end{aligned}$$

Hence

$$|c_n - L| < \varepsilon$$

✓

Example. Show $\lim_{n\to\infty} n^{\frac{1}{n}} = 1$

By sandwich theorem, we first have $n^{\frac{1}{n}} \geq 1$ for all $n \geq 1$. Hence we have found a lower bound for all terms with $n > 1$.

We then seek an upper bound that goes to 1. Define a helper $h_n = n^{\frac{1}{n}} - 1$, then $n^{\frac{1}{n}} = 1 + h_n$. We want to find an upper bound of h_n that goes to 0. Write

$$\begin{aligned} n^{\frac{1}{n}} &= 1 + h_n \\ \left(n^{\frac{1}{n}}\right)^n &= (1+h_n)^n \\ &= \sum_{k=0}^{n} \binom{n}{k} h_n^k \\ &\geq \binom{n}{2} h_n^2 \\ &= \frac{n(n-1)}{2} h_n^2 \\ \frac{2n}{n(n-1)} &\geq h_n^2 \\ \sqrt{\frac{2}{n-1}} &\geq h_n \end{aligned}$$

Hence we can apply the sandwich theorem and say that $n^{\frac{1}{n}} \to 1$.

Definition 3.9 (Divergence)**.** We say $\{a_n\}$ diverges to ∞ if $\forall M > 0$, $\exists N > 0$ s.t. $a_n > M, \forall n > N$, denoted

$$\lim_{n\to\infty} a_n = \infty$$

Example. $\{0, 2, 0, 4, \ldots\}$ does not diverge to ∞.

3.2 Cauchy sequences and monotone sequences

Definition 3.10 (Cauchy sequence)**.** $\{a_n\}$ is a **Cauchy sequence** if $\forall \varepsilon > 0$, $\exists N > 0$ s.t. $|a_n - a_m| < \varepsilon, \forall n, m > N$.

Theorem 3.11 (Cauchy implies boundedness)**.** Every Cauchy sequence is bounded.

Proof. Fix $\{a_n\}$ a Cauchy sequence. Fix $\varepsilon = 1$, by definition, $\exists N_0 > 1$ s.t. $\forall n, m > N_0$, $|a_n - a_m| < 1$. Hence, $a_n < |a_{N_0+1}| + 1, \forall n > N_0$.

Also, $\{a_n\}_{1 \leq n \leq N_0}$ is finite and a maximum exists.

Hence $M \coloneqq \max\{a_1, \ldots a_{N_0}, |a_{N_0+1}|\}$ bounds the whole sequence. ✓

Theorem 3.12 (Convergence imply Cauchyness)**.** Every convergent sequence is Cauchy.

Proof. Fix sequence $\{a_n\}$, fix $\varepsilon > 0$. Let limit be L. Since a_n converges, $\exists N > 0$ s.t.

$$\begin{aligned} &|a_n - L| < \frac{\varepsilon}{2}, \forall n > N \\ \implies &\forall n, m > N, |a_n - L| < \frac{\varepsilon}{2}, |a_m - L| < \frac{\varepsilon}{2} \\ \implies &|a_n - a_m| = |(a_n - L) - (a_m - L)| \leq |a_n - L| + |a_m - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

✓

Theorem 3.13 (Cauchy Convergence Criterion)**.** If $\{a_n\} \in \mathbb{R}$ is Cauchy then it converges, and the limit is in $\mathbb{R}$.

Proof. *We use Bolzano-Weierstrass to show existence of limit.*

Consider two cases. $\{a_n\}$ is either finite or infinite.

- $\{a_n\}$ finite. Let S dnote the set of values $\{s_1, \ldots s_k\}$. Let $\varepsilon = \frac{1}{2}\min\{|s_i - s_j|\}, i \neq j, 1 \leq i, j \leq k$.

  Since $\{a_n\}$ is Cauchy, $\exists N > 0$ s.t.

  $$|a_n - a_m| < \varepsilon, \forall n, m > N \implies \exists i \text{ s.t. } a_n = s_i, \forall n > N$$

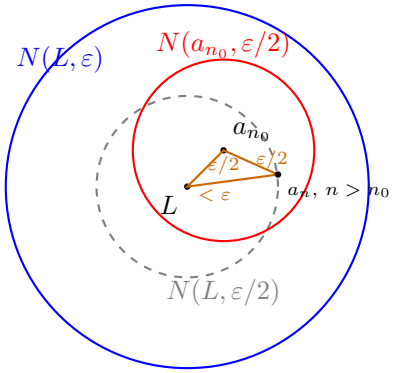
- $\{a_n\}$ infinite. We have shown above that Cauchyness implies boundedness. By BW, $\exists L \in \mathbb{R}$ that is an accumulation point.

  WTS $\lim_{n\to\infty} a_n = L$. Fix $\varepsilon > 0$, we seek N s.t. $|a_n - L| < \varepsilon, \forall n > N$.

  Since $\{a_n\}$ Cauchy, $\exists N_0$ s.t. $|a_n - a_m| < \frac{\varepsilon}{2}, \forall n, m > N$. If suffices to show that at least one $a_{n_0} \in N\left(L, \frac{\varepsilon}{2}\right), n_0 > N_0$.
  Since L is an accumulation point, $N^*\left(L, \frac{\varepsilon}{2}\right)$ contains infinitely many points, so in particular we may pick $n_0 > N_0$ with $a_{n_0} \in N\left(L, \frac{\varepsilon}{2}\right)$. Then for $n > n_0$,

$$|a_n - L| \leq |a_n - a_{n_0}| + |a_{n_0} - L| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,$$

  i.e. $N\left(a_{n_0}, \frac{\varepsilon}{2}\right) \subseteq N(L, \varepsilon)$ and every later term lies in it.



✓

Example. We can show divergence of Harmonic series by showing that it is not Cauchy.

$$a_n = 1 + \frac{1}{2} + \ldots + \frac{1}{n}$$

i.e. WTS $\exists \varepsilon > 0, \forall N > 0, \exists m > n > N$ s.t. $|a_m - a_n| > \varepsilon$. Take $\varepsilon = \frac{1}{4}$, $m = 2n$, then

$$\begin{aligned} a_m - a_n &= \frac{1}{n+1} + \frac{1}{n+2} + \ldots \frac{1}{m} \\ &\geq \frac{m-n}{m} \\ &= 1 - \frac{n}{m} \\ &= \frac{1}{2} \end{aligned}$$

Hence the harmonic series is not cauchy and therefore does not converge.

Definition 3.14 (Monotone sequence)**.**
1. $\{a_n\}$ is **increasing** if $a_n \leq a_{n+1}, \forall n$
2. $\{a_n\}$ is **decreasing** if $a_n \geq a_{n+1}, \forall n$

If $\{a_n\}$ is either increasing or decreasing, we say it is **monotone**.

Equivalently, we can define strictly increasing, strictly decreasing and strictly monotone.

Theorem 3.15 (Monotone Convergence Theorem)**.** If a sequence $\{a_n\}$ is monotone and bounded, then it converges.

Corollary 3.16 (Alternative statement of MCT)**.** Given a monotone sequence, it is bounded if and only if it converges.

Proof of MCT. WTS that given a monotone sequence, it converges if and only it is bounded.

($\implies$): convergence implies boundedness always, without requiring monotonicity.

($\impliedby$): Let $\{a_n\}$ be a monotone and bounded sequence, WTS it converges.

WLOG, assume $\{a_n\}$ increasing. Since $\{a_n\}$ bounded, by CA, $\sup\{a_n\}$ exists, say L. WTS $\lim_{n\to\infty} a_n = L$.

Fix $\varepsilon > 0<$ by definition of supremum, $\exists n_0$ s.t. $a_{n_0} > L - \varepsilon$. Hence

$$\begin{aligned} & L - \varepsilon < a_n \leq L, \forall n > n_0 \\ \implies & |a_n - L| < \varepsilon, \forall n > n_0 \end{aligned}$$

✓

Remark.

1. An increasing sequence converges $\iff$ bounded from above.
2. A decreasing sequence converges $\iff$ bounded from below.

Example. *We can use MCT to determine the convergence of a sequence defined recursively, i.e. we prove convergence by showing boundedness and monotonicitiy*

For example, $a_1 = 5$, $a_{n+1} = \frac{1}{2}\left(a_n + \frac{5}{a_n}\right)$. We can show this is bounded by noting $a_n > 0$ and therefore

$$\begin{aligned} a_{n+1} &= \frac{1}{2}\left(\sqrt{a_n}^2 + \sqrt{\frac{5}{a_n}}^2\right) \\ &\geq \sqrt{a_n}\sqrt{\frac{5}{a_n}} = \sqrt{5} \end{aligned}$$

Proof that interval are connected .

Recall theorem 2.21. WTS that if $S \subset \mathbb{R}$ is an interval then S is connected.

Suppose S disconnected. By definition, $\exists A, B$ open and nonempty such that $S \subset A \cup B$, $A \cap B = \emptyset$, $S \cap A \neq \emptyset$, $S \cap B \neq \emptyset$.

Let $x \in S \cap A, y \in S \cap B$. WLOG, say $x < y$. Let $x_1 = x, y_1 = y$. We seek sequences $\{x_n\} \subset A \cap S$, $\{y_n\} \subset A \cap S$, such that

- x_n is increasing
- y_n is decreasing
- $|x_n - y_n| \to 0$ as $n \to \infty$

To construct such a sequence, take $z_{k+1} = \frac{x_k + y_k}{2}$.

- if $z_{k+1} \in A, x_{k+1} = z_{k+1}, y_{k+1} = y_k$
- if $z_{k+1} \in B, x_{k+1} = x_k, y_{k+1} = z_{k+1}$

Then

$$y_i - x_i = \frac{y_1 - x_1}{2^{i-1}}$$

Since $\{x_n\}$ bounded and increasing, $\exists a \in \mathbb{R}$ s.t. $x_n \to a$. Likewise $\exists b \in \mathbb{R}$ s.t. $y_n \to b$. Then

$$b - a = \lim_{n\to\infty} y_n - \lim_{n\to\infty} x_n = \lim_{n\to\infty} (y_n - x_n) = \lim_{n\to\infty} 2^{-(n-1)}(y_1 = x_1) = 0$$

Hence $b = a$.

We know that $a \in (S \cap A) \setminus (S \cap B)$ or $a \in (S \cap B) \setminus (S \cap A)$. Suppose $a \in S \cap A$ but $a \notin B$, by openess of A, there is some $N(a, \varepsilon) \subset A$. Since A, B disjoint, $N(a, \varepsilon) \subset B^c$. However, $y_n \to a$, hence $\exists N > 0$ s.t. $\forall n > N$, $y_n \in N(a, \varepsilon)$, which is a contradiction. ✓

3.3 Subsequences

Definition 3.17 (Subsequence)**.** Let $\{a_n\} \in \mathbb{R}$ and $\{n_k\}_k$ be a striclty increasing sequence in $\mathbb{N}$, i.e. $n_k < n_{k+1}, \forall k$, then $\{a_{n_k}\}_k$ is a subsequence of $\{a_n\}$

Example. Take $\{1, -1, 1, -1, \ldots\}$,

- $n_k = 2k - 1, \{n_k\}_k = \{1, 3, 5, \ldots\}$, $\{a_{n_k}\}_k = \{1, 1, 1, \ldots\}$
- $n_k = 2k, \{n_k\}_k = \{2, 4, 6, \ldots\}$, $\{a_{n_k}\}_k = \{-1, -1, -1, \ldots\}$

Theorem 3.18 (Subsequence of convergent sequence converges)**.** If $\lim_{n\to\infty} a_n = L \in \mathbb{R}$, or $L = \pm\infty$ then for any subsequence $\{a_{n_k}\}_k$ of $\{a_n\}$,

$$\lim_{k\to\infty} a_{n_k} = L$$

Proof. Fix $\varepsilon > 0$, we seek $K > 0$ s.t. $|a_{n_k} - L| < \varepsilon$, $\forall k > K$. Since a_n converges,

$$\lim_{n\to\infty} a_n = L \implies \exists N > 0 \text{ s.t } |a_n - L| < \varepsilon, \forall n > N$$

Also, $\forall k > N, n_k \geq k \geq N$, hence $|a_{n_k} - L| < \varepsilon$, hence

$$\lim_{k\to\infty} a_{n_k = L}$$

✓

Corollary 3.19 (Convergent subsequences whose limits disagree imply divergence)**.** If $\{a_n\}$ has two subsequences that converge to different values then $\{a_n\}$ diverges and does not converge to $\pm\infty$.

Theorem 3.20 (Bdd divergent sequences have at least two convergent subsequences)**.** If $\{a_n\}$ bounded and divergent, then there exist two subsequences of $\{a_n\}$ that converge to different values.

Theorem 3.21. Every bounded sequence $\{a_n\}$ has a convergent subsequence.

Proof of 3.21 $\implies$ 3.20.
Fix a bounded and divergent sequence $\{a_n\}_n$ 3.21, $\exists\{a_{N_k}\}_k$ and $L_1 \in \mathbb{R}$ s.t. $a_{N_K} \to L$. By definition of divergence, we know that $\{a_n\}$ does not converge, i.e. fix $\varepsilon > 0$,

$$\begin{aligned} &\forall N > 0, \exists n > N \text{ s.t. } |a_n - L_1| > \varepsilon \\ \implies &\exists\{a_{n_l}\}_l \text{ s.t. } |a_{n_l} - L_1| > \varepsilon, \forall l \in \mathbb{N} \end{aligned}$$

Since $\{a_{n_l}\}_l$ bounded, it has a convergent subsequence by 3.21. WLOG, assume $\{a_{n_l}\}$ converges to L_2.

Since $|a_{n_l} - L_2| > \varepsilon, \forall l \in \mathbb{N}$, $|L_2 - L_1| \geq \varepsilon > 0$. ✓

Proof of 3.21.
Consider 2 cases

- Case 1: $\{a_n\}$ finite Then at least one value is attained infinitely many times, take a subsequence consisting only of that value, and we have found a convergent subsequence.
- Case 2: $\{a_n\}$ infinite By BW, the set of values attained by $\{a_n\}$ has an accumulation point say L.

  We want to construct $\{a_{\lfloor n_k \rfloor}\}$ converging to L.
  - Let $\varepsilon_1 \coloneqq 1$, and $a_{n_1} \in N^*(L, \varepsilon_1) \cap \{a_n\}$.
  - Let $\varepsilon_2 = \frac{1}{2}$. Since $N^*(L, \varepsilon_2) \cap \{a_n\}_{n \geq n_1+1}$ has infinitely many points, $\exists n_2 > n$ s.t. $a_{n_2} \in N^*(L, \varepsilon_2)$.
  - Repeat this process, if $a_{n_1}, a_{n_2}, \ldots a_{n_k}$ chosen, let $\varepsilon_{k+1} = \frac{1}{k+1}$ and pick

    $$a_{n_{k+1}} \in N^*(L, \varepsilon_{k+1}) \cap \{a_n\}_{n \geq n_K + 1}$$

    , and we have constructed a subsequence $\{a_{n_k}\}$ such that $a_{n_k} \to L$

✓

Definition 3.22 (Subsequential limits)**.** Let $\{a_n\} \subset \mathbb{R}$, the set of subsequential limits of $\{a_n\}$ is the set of $x \in \mathbb{R} \cup \{\pm\infty\}$ such that some subsequence $\{a_{n_k}\}$ converges to x.

Remark. $\{a_n\}$ bounded if and only if its set of subsequential limits is a nonempty and bounded subset of $\mathbb{R}$.

Definition 3.23 (Extended reals)**.** In the extended real number system $R \cup \{\pm\infty\}$,

- we have order

  $$-\infty < x < \infty, \forall x$$

- we do not have a field

- we have the following conventions, $\forall x \in \mathbb{R}$

$$x + \infty = \infty$$
$$x - \infty = -\infty$$
$$\frac{x}{+\infty} = \frac{x}{-\infty} = 0$$
$$x > 0 \implies x \times (\pm\infty) = \pm\infty$$
$$x < 0 \implies x \times (\pm\infty) = \mp\infty$$

Definition 3.24 (Limit superior, limit inferior)**.** The limit superior of $\{a_n\}$ is defined as the supremum of the set of subsequential limit, denoted as

$$\limsup_{n\to\infty} a_n \in \mathbb{R} \cup \{\pm\infty\}$$

Similarly, the limit inferior is defined as the infimum of the set of subsequential limits, denoted $\liminf_{n\to\infty} a_n$.

Remark. The limit inferiors and superiors always exists for any sequence in $\mathbb{R} \cup \{\pm\infty\}$.

- $\{a_n\}$ bounded from above $\iff \limsup_{n\to\infty} a_n \in \mathbb{R} \cup \{-\infty\}$. Otherwise $\limsup_{n\to\infty} a_n = \infty$.
- $\{a_n\}$ bounded from bleow $\iff \liminf_{n\to\infty} a_n \in \mathbb{R} \cup \{\infty\}$. Otherwise $\liminf n \to \infty a_n = -\infty$.
- $\{a_n\}$ bounded $\implies \liminf a_n, \limsup a_n \in \mathbb{R}$.

Proposition 3.25.

1. $\liminf_{n\to\infty} a_n \le \limsup_{n\to\infty} a_n$
2. A sequence converges if and only if its limsup and liminf agree

$$\lim_{n\to\infty} a_n = L \in \mathbb{R} \cup \{\pm\infty\} \iff \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = L$$

Proof. **WTS** $\lim_{n\to\infty} a_n = L \implies \liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = L$ Since every subsequence of a convergent sequence converges to the same limit.

WTS $\liminf_{n\to\infty} a_n = \limsup_{n\to\infty} a_n = L \implies \lim_{n\to\infty} a_n = L$ ATC otherwise, i.e.

$$\lim_{n\to\infty} a_n \neq L \implies \qquad \exists \{a_{n_k}\} \text{ s.t. } \exists \varepsilon_0, |a_{n_k} - L| > \varepsilon_0, \forall k \in \mathbb{N}$$

Since $\liminf a_n = \limsup a_n = L$, $\{a_n\}$ must be bounded. Hence $\{a_{n_k}\}$ also bounded. Hence $\{a_{n_k}\}$ has a convergent subsequence, which we still denote $\{a_{n_k}\}$. Since this is also a subsequence of the original $\{a_n\}$, we know that $\lim_{k\to\infty} a_{n_k} = L$. However, this leads to a contradiction because take $\varepsilon = \frac{\varepsilon_0}{2}$, then $\forall k, |a_{n_k} - L| > \varepsilon_0 > \varepsilon$. ✓

Proposition 3.26. If $\{a_n\}$ is bounded then its set of subsequential limits is closed.

Proof. *The idea is to take any accumulation point in the set of subsequential limits of a_n, and show that some subsequence indeed converges to that accumulation point.*

If $\{a_n\}$ bounded, then all of its subsequential limits are in $\mathbb{R}$. Let E denote the set of subsequential limits. WTS that E contains all of its accumulation points, i.e. $\forall p \in E', p \in E$.

Now we want to construct a subsequence that converges to p

- First choose $p_1 \in N^*(p, 1) \cap E$. Let $\varepsilon_1 = 1 - |p - p_1| > 0$. $\exists a_{n_1}$ s.t. $|a_{n_1} - p_1| < \varepsilon_1 \implies |a_{n_1} - p| < 1$.
- $\vdots$
- Suppose $a_{n_1}, \dots a_{n_{k-1}}$ are found, s.t. $n_1 < n_2 < \dots n_{k-1}$, choose $p_k \in N^*(p, \frac{1}{k}) \cap E$. Let $\varepsilon_k = \frac{1}{k} - |p - p_k| > 0$. $\exists a_{n_k}$ s.t. $n_k > n_{k-1}$ and $|a_{n_k} - p_k| < \varepsilon_k$. Hence $|a_{k_k} - p| < \frac{1}{k}$.

Therefore $a_{n_k} \to p$ and $p \in E$. ✓

Theorem 3.27. If $\{a_n\}$ is bounded and $x \in \mathbb{R}$, TFAE

1. $\limsup_{n\to\infty} a_n = x$
2. (a) $\forall \varepsilon > 0, \exists N > 0$ s.t. $\forall n > N$, $a_n < x + \varepsilon$ AND
   (b) $\forall \varepsilon > 0, \forall N > 0$, $\exists n > N$, s.t. $a_n > x - \varepsilon$
3. (a) There is $\{a_{n_k}\}$ that converges to x AND
   (b) If $\{a_{n_j}\}$ converges, then $\lim_{j\to\infty} a_j \le x$

4. Let $b_n = \sup_{m \geq n} a_m$, then b_n converges and $\lim_{n\to\infty} b_n = x$

Proof of 3.27, In general, let E denote the set of subsequential limits of a_n.

- (1) $\iff$ (2)
  1. (1) $\implies$ (2)
     - (1) $\implies$ (2a): ATC otherwise, then $\exists$ infinitely many terms where $a_n \geq x + \varepsilon$. Then they form a subsequence $\{a_{n_k}\}$ which is also bounded. Hence this subsequence itself has a convergent subsequence converging to some limit $L \geq x + \varepsilon > x$. Hence contradicting the assumption that x is a supremum.
     - (1) $\implies$ (2b) ATC otherwise, then only finitely many terms $a_n > x - \varepsilon$. Hence, $\exists N > 0$ s.t. $\forall n > N$, $a_n \leq x - \varepsilon$. Hence all subsequential limits must be less than $x - \varepsilon$. This contradicts the assumption that x is a minimal upper bound.
  2. (1) $\impliedby$ (2): Fix any $\varepsilon > 0$, $\exists N$ s.t. $\forall n > N$, $a_n < x + \varepsilon$. Hence any subsequential limit is smaller than $x + \varepsilon$. Therefore $\limsup a_n \leq x + \varepsilon$. Since ε is arbitrary, $\limsup a_n \leq x$.
- (1) $\iff$ (3)
  1. (1) $\implies$ (3) Since the set of subsequential limits is closed, the supremum is in E. Hence some subsequence converges to it. By definition of supremum, $x \geq p, \forall p \in E$.
  2. (1) $\impliedby$ (3): by definition
- (1) $\iff$ (4)
  1. (1) $\implies$ (4): Since $\{a_n\}$ bounded, $\{\sup_{m\geq n} a_m\}_n$ is also bounded by the same bounds. Also note

$$\sup_{m\geq n+1} a_m \leq \sup_{m\geq n} a_m$$

     Hence the sequence is decreasing. By MCT

$$\lim_{n\to\infty} \sup_{m\geq n} a_m = L \in \mathbb{R}$$

     It suffices to show that $L = x$. Fix $\varepsilon > 0$.
     - By (2a): $\exists N > 0$ s.t. $\forall n > N$ $a_n < x + \frac{\varepsilon}{2}$. Hence

$$b_n = \sup_{m\geq n} a_m \leq x + \frac{\varepsilon}{2} < x + \varepsilon$$

       Hence $L = \lim_{n\to\infty} b_n \leq x + \varepsilon \implies L \leq x$
     - By (2b): $\forall n > 0$,

$$b_n = \sup_{m\geq n} a_m > x - \varepsilon$$

       Hence $L = \lim_{n\to\infty} b_n \geq x - \varepsilon \implies L \geq x$.
  2. (1) $\impliedby$ (4): Prove by showing (4) $\implies$ (3).

     We pick
     - n_1 s.t. $a_{n_1} \geq \sup_{m\geq 1} a_m - 1$
     - n_2 s.t. $n_2 > n_1$ and $a_{n_2} \geq \sup_{m\geq n_1+1} a_m - \frac{1}{2}$.
     - $\vdots$
     - n_k s.t. $n_k > n_{k-1}$ s.t. $a_{n_k} \geq \sup_{m\geq n_{k-1}+1} a_m - \frac{1}{k}$.

     Let $L = \lim_{n\to\infty} \sup_{m\geq n} a_m$.

     WTS $\lim_{n\to\infty} a_{n_k} = L$. Fix $\varepsilon > 0$. Choose $N > 0$ s.t. $\forall n > N$,

$$\left|\sup_{m\geq n} a_m - L\right| < \frac{\varepsilon}{2}$$

     Then take $N_0 = \max\{N + 1, \frac{2}{\varepsilon}\}$, one has $\forall k > N_0$,

$$\begin{aligned} |a_{n_k} - L| &\leq \left|a_{n_k} - \sup_{m\geq n_{k-1}+1} a_m\right| + \left|\sup_{m\geq n_{k-1}+1} - L\right| \\ &\leq \frac{1}{k} + \frac{\varepsilon}{2} \\ &\leq \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

     Hence we have shown (3a).

Let $\{a_{n_k}\}$ be any convergent subsequence. of $\{a_n\}$. We have

$$a_{n_k} \leq \sup_{m \geq n_k} a_m, \forall k$$
$$\implies \lim_{k \to \infty} a_{n_k} \leq \lim_{k \to \infty} \sup_{m \geq n_k} a_m = L$$

✓

4 Limits and Continuity of Functions

4.1 Limits of Functions

Definition 4.1 (Limits of functions)**.** Let $f : D \to \mathbb{R}, D \subset \mathbb{R}$, and c be an accumulation point of D. We say $L \in \mathbb{R}$ is the limit of f at c if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|f(x) - L| < \varepsilon$ whenever $0 < |x - c| < \delta$ and $x \in D$, denoted

$$\lim_{x \to c} f(x) = L$$

Remark.

1. We define limits of functions only on accumulation points in the domain.
2. Note that $0 < |x - c|$, i.e. we care about $x \in N^*(c, \delta)$. We do not need $f(c)$ to be defined. When it is defined, $\lim_{x \to c} f(x)$ has nothing to do with $f(c)$.
3. The limit of a function at c is unique.

Example. Show $\lim_{x \to 2} x^2 = 4$. **Add in later**.

Theorem 4.2 (Equivalence between limit of function and limit of a sequence)**.**
Let $f : D \to \mathbb{R}, c \in D'$, TFAE

1. $\lim_{x \to c} f(x) = L$
2. For every $\{a_n\} \subset D$ s.t. $a_n \to c$ but $a_n \neq c, \forall n$, we have

$$\lim_{n \to \infty} f(a_n) = L$$

Remark.

- This result allows us to work with limits of sequences whenever we want to talk about limits of continuous functions
- We can make the above theorem weaker by only requiring $a_n \neq c \ \forall n > N$ for some N.

Example. $\lim_{x \to \infty} \sin\left(\frac{1}{x}\right)$ does not have a limit.

Proof. To show (1) $\implies$ (2): Assume $\lim_{x \to c} f(x) = L$, Fix $\{a_n\} \subset D$ s.t. $a_n \to c, a_n \neq c, \forall n \in \mathbb{N}$ WTS $\lim_{n \to \infty} f(a_n) = L$.

By definition, $\lim_{x \to c} f(x) = L \implies \exists \delta$ s.t. $0 < |x - c| < \delta \implies |f(x) - L| < \varepsilon$. Since $a_n \to c \iff \lim_{n \to \infty} a_n = c$ and $a_n \neq c, \forall n$, $\exists N > 0$ s.t. $\forall n > N$

$$0 < |a_n - c| < \delta \implies |f(a_n) - L| < \delta$$

To show (2) $\implies$ (1): Assume $f(x)$ does not converge to L as $x \to c$, $\exists \varepsilon > 0$ s.t. $\forall \delta > 0$, $\exists x \in N^*(c, \delta) \cap D$ s.t. $|f(x) - L| \geq \varepsilon$.

- Let $\delta_1 = 1, a_1 \in N^*(c, \delta_1) \cap D$ s.t. $|f(a_1) - L| \geq \varepsilon$
- Let $\delta_2 = \frac{1}{2}, a_2 \in N^*(c, \delta_2) \cap D$ s.t. $|f(a_2) - L| \geq \varepsilon$
- $\vdots$
- $\delta_n = \frac{1}{n}, a_n \in N^*(c, \delta_n) \cap D$ s.t. $|f(a_n) - L| \geq \varepsilon$

Since a_n satisfies $a_n \to c, a_n \neq c$, by assumption, $\lim_{n \to \infty} f(a_n) = L$. This contradicts $|f(a_n) - L| \geq \varepsilon$. ✓

Corollary 4.3. Let $f : D \to \mathbb{R}, c \in D'$, TFAE

1. f doesn't have a limit at c
2. There exists a sequence $\{a_n\} \subset D$ that converges to c with $a_n \neq c, \forall n$ s.t. $\{f(a_n)\}$ diverges.

Remark. This is very useful for proving that a limit DNE.

Proposition 4.4 (Algebraic operations of limits of functions)**.** Let $f : D \to \mathbb{R}, c \in D'$. Suppose $\lim_{x \to c} f(x) = L$, $\lim_{x \to c} g(x) = M$. Then

1. $\lim_{x \to c}(f + g)(x) = L + M$.
2. $\lim_{x \to c}(f \cdot g)(x) = LM$.
3. $\lim_{x \to c} kf(x) = kL, \forall k \in \mathbb{R}$.

4. If $g(x) \neq 0, \forall x \in D$, and $M \neq 0$, then
$$\lim_{x\to c} \frac{f}{g}(x) = \frac{L}{M}$$

Definition 4.5 (One sided limits)**.**
1. We say that $\lim_{x\to c^-} f(x) = L$ if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $c - \delta < x < c \implies |f(x) - L| < \varepsilon$
2. We say that $\lim_{x\to c^+} f(x) = L$ if $\forall \varepsilon > 0$, $\exists \delta > 0$ s.t. $c < x < c + \delta \implies |f(x) - L| < \varepsilon$

Proposition 4.6 (Equivalence of one-sided and two-sided limits)**.**
$$\lim_{x\to c} f(x) = L \iff \lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L$$

4.2 Continuous functions

Definition 4.7 (Continuity)**.** Let $f : D \to \mathbb{R}$
1. (**Continuity at a point**) Let $c \in D$, we say f is **continuous at** c if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $|f(x) - f(c)| < \varepsilon$ whenever $|x - c| < \delta$.
2. (**Continutiy over a set**): Let $S \subset D$ if $\forall c \in S$, f is continuous at c, then we say that f is **contiuous on** S.
3. (**Continuous function**): if f is continuous on D, then we say f is a **continuous function**.

Remark.
1. Compared to the definition of the limit of a function at a point, we now require $c \in D$ instead of $c \in D'$.
2. We now allow $x = c$.
3. We cannot immediately apply the previous definition of limit of a function at point to define continuity because we don't know that c is an accumulation point.
4. Depending on whether c is an accumulation point or an isolated point, we have different conditions that are required for the continuity of f at c. Refer to the following two propositions.

Proposition 4.8 (Any function is continuous at isolated point)**.** If c is an isolated point of D, then f is always continuous at c.

Proof. c isolated $\implies \exists \delta_0$ s.t. $N^*(c, \delta_0) \cap D = \emptyset$. Fix any $\varepsilon > 0$, pick $\delta = \delta_0$. Then,
$$x \in N(c, \delta) \implies x = c \implies |f(x) - f(c)| = 0 < \varepsilon$$
✓

Proposition 4.9. If c is an accumulation point of D, then f is continuous at c if and only if $\lim_{x\to c} f(x) = f(c)$.

Proof. By definition, **exercise**. ✓

Proposition 4.10 (Equivalent characterization of continuity)**.** f is continuous at c if any only if for any neighborhood of $f(c)$, say $V = N(f(c), \varepsilon), \forall \varepsilon$, there exists a neighborhood $U = N(c, \delta)$ such that $f(U \cap D) = V$.

Theorem 4.11. Let $f : D \to \mathbb{R}, c \in D$, TFAE
1. f is continuous at c
2. If $\{a_n\} \subset D$ s.t. $a_n \to c$, then
$$\lim_{n\to\infty} f(a_n) = f(c)$$

Proof. **revisit this proof**

c is either an isolated point or an accumulation point.
1. Case 1: c isolated. We want to show that (1) and (2) are always true.
   - We already know that (1) is always true, any f is continuous at c if c isolated.
   - WTS (2). Fix any $\{a_n\}$ converging to c. Since c isolated, $\exists \varepsilon > 0$ s.t. $N^*(c, \varepsilon) \cap D = \emptyset$. Since $a_n \to c, \exists N$ s.t. $\forall n > N$, $|a_n - c| < \varepsilon \implies a_n = c \implies f(a_n) = f(c)$.
2. Case 2: c accumulation point

- (1) $\implies$ (2):

$$f\text{continuous at } c \implies \lim_{x\to c} f(x) = f(c) \text{ by definition}$$
$$\implies \{a_n\} \subset D, a_n \neq c, a_n \to c \implies \lim_{n\to\infty} f(a_n) = f(c)$$

  ATC otherwise, i.e. $\{a_n\} \subset D, a_n \to c, \lim_{n\to\infty} f(a_n) \neq f(c)$ then $\exists \varepsilon > 0$ and some subsequence $\{a_{n_k}\}$ such that $a_{n_k} \to c$ but $|f(a_{n_k}) - f(c)| > \varepsilon$. Hence $a_{n_k} \neq c, \forall k$. However, this contradicts with the result.
- (2) $\implies$ (1): Assume (2), then we immediately have

$$\{a_n\} \subset D, a_n \neq c, a_n \to c \implies \lim_{n\to\infty} f(a_n) = f(c)$$
$$\implies \lim_{x\to c} f(x) = f(c)$$
$$\implies f \text{ continuous at } c$$

✓

Theorem 4.12. $f, g : D \to \mathbb{R}, c \in D$. Suppose f, g both continuous at c. Then the following functions are also continuous at c:

1. $f + g$
2. fg
3. kf
4. $\frac{f}{g}$, if $g(c) \neq 0$

Remark. The assumption that $\exists \varepsilon > 0$ s.t. $\forall x \in N(c, \varepsilon), g(x) \neq 0$ is necessary and sufficient condition for $g(c) \neq 0$.

Proof. We prove using the previous theorem, and the analogous result for limits. ✓

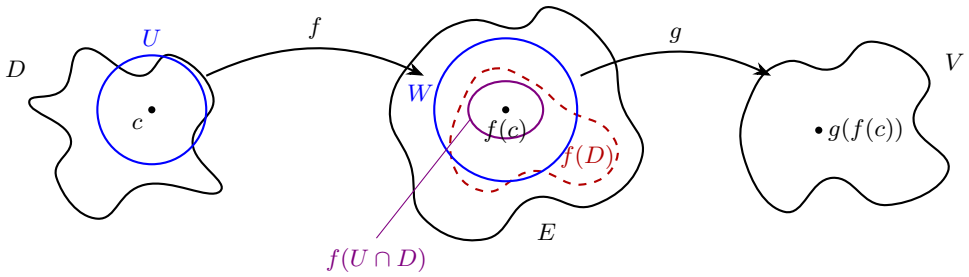
Theorem 4.13. Let $f : D \to \mathbb{R}, g : E \to \mathbb{R}$ s.t. $f(D) \subset E$, suppose f continuous at $c \in D$, g continuous at $f(c) \in E$, then $g \circ f : D \to \mathbb{R}$ is continuous at c.

Proof. Fix a neighborhood V of $g \circ f(c)$, we seek a neighborhood U such that $g \circ f(U \cap D) \subset V$.

Since g continuous at $f(c)$, $\exists$ a neighborhood W of $f(c)$ such that $g(W \cap E) \subset V$.

Since f continuous at c, $\exists$ a neoighboord U of c such that $f(U \cap D) \subset W$. ✓

Remark. $f(D) \subset E \implies f(U \cap D) \subset W \cap E \implies g \circ f(U \cap D) \subset g(W \cap E) \subset V$.



Theorem 4.14 (Limit passes through continuous functions)**.**
Let $f : D \to \mathbb{R}, g : E \to \mathbb{R}$, $f(D) \subset E$, suppose $\lim_{x\to c} f(x) = L \subset E$ and g is continuous at L, then

$$\lim_{x\to c} g(f(x)) = g\left(\lim_{x\to c} f(x)\right) = g(L)$$

Example. If however, we only have $\lim_{x\to c} f(x) = L$, $\lim_{y\to L} g(y) = M$, then in general we **do not have** $\lim_{x\to c} g(f(x)) = M$.

For example, take $f(x) = L$ a constant function, then $\lim_{x\to c} g(f(x)) = g(L)$, which is not necessarily equal to $\lim_{y\to L} g(y)$, unless we know that g is continous at L.

Proof. Fix $\varepsilon > 0$. WTS $\exists \delta > 0$ s.t. if $0 < |x - c| < \delta$ and $x \in D$, then $|g(f(x)) - g(L)| < \varepsilon$.

By definition of continituity of g at L, for ε, $\exists \eta > 0$ s.t. $|y - L| < \eta, y \in E \implies |g(y) - g(L)| < \varepsilon$

By definition of limit of f at c, since $\lim_{x\to c} f(x) = L$, $\exists \delta > 0$ s.t. $0 < |x - c| < \delta, x \in D, \implies |f(x) - L| < \eta$.

Since $f(D) \subset E$, $0 < |x - c| < \delta, x \in D \implies |f(x) - L| < \eta, f(x) \in E \implies |g(f(x)) - g(L)| < \varepsilon$. ✓

Example. $f(x) = x^2, D = (-1, 1), f(D) = [0, 1)$

Remark. Continuous functions do not necessarily preserve openess.

Theorem 4.15 (Preimage of open set in $\mathbb{R}$ is open). Let f be a function on $\mathbb{R}$, TFAE
- f is continuous on $\mathbb{R}$.
- For any open $V \subset \mathbb{R}$, $f^{-1}(V)$ is open.

Remark. This can be made stronger into the following theorem.

Theorem 4.16 (Preimage of open is open). Let $f : D \to \mathbb{R}$, TFAE
1. f is continuous on D
2. Any open image has a preimage that is the intersection of an open set and the domain.

$$\forall V \subset \mathbb{R}, V \text{ open}, \exists U \subset \mathbb{R} \text{ s.t. } f(U \cap D) = V, \text{ i.e. } f^{-1} = U \cap D.$$

Remark. We want to say that any open image has an open pre-image, but that is not always true. We can only say $f^{-1}(V) = U \cap D$ for some open U.

Corollary 4.17. Let $f : D \to \mathbb{R}$, TFAE
1. f is continuous on D
2. for any closed $V \subset \mathbb{R}$, $\exists U \subset \mathbb{R}$, U closed, s.t. $f^{-1}(V) = D \cap U$.

Proof. We prove a special case where $D = \mathbb{R}$.

(1) $\implies$ (2): Assume f continous on D, let V be open. WTS $f^{-1}(V)$ open.

Fix any $c \in f^{-1}(V)$. WTS $\exists \varepsilon > 0$ s.t $N(c, \varepsilon) \subset f^{-1}(V)$.

Since $c \in f^{-1}(V), f(c) \in V$. Since V open, $\exists$ a neighborhood of $f(c)$ say U s.t. $U \subset V$.

Since f continuous at c, by definition, $\exists \varepsilon > 0$ s.t. $f(N(c, \varepsilon)) \subset U \subset V$. Hence $N(c, \varepsilon) \subset f^{-1}(V)$.

(2) $\implies$ (1): WTS f continuous at any $c \in \mathbb{R}$. Let V be any neighborhood of $f(c)$. WTS $\exists$ a neighborhood of c, say U s.t. $f(U) \subset V$. By assumption, we know $f^{-1}(V)$ is open. Since $c \in f^{-1}(V)$, c must be an interior point of $f^{-1}(V)$. So there exists a neighborhood of c say U s.t. $U \subset f^{-1}(V)$, i.e. $f(U) \subset V$. ✓

4.3 Properties of continuous functions: compactness

Remark. Continuity does not preserve closedness.

Example. Consider $D = [1, \infty), f(x) = \frac{1}{x}$. Then the domain is closed, but the image $f(D) = (0, 1]$ and is not closed.

Remark. Continuity does not preserve boundedness.

Example. Consider $D = [0, 1], f(x) = \frac{1}{x}$. Then the domain is bounded, but the image $f(D) = [1, \infty)$ and is not bounded.

Theorem 4.18 (Continuity preserves compactness). Let $D \subset \mathbb{R}$ be compact, and $f : D \to \mathbb{R}$ be continuous. Then, $f(D)$ is compact.

Proof. Let $\mathcal{F} = \{O_\alpha\}_{\alpha \in \Delta}$ be an open cover for $f(D)$. WTS there exists a finite subcover for $f(D)$.

By theorem, each $\forall O_\alpha$, $\exists H_\alpha$ open s.t. $f^{-1}(O_\alpha) = H_\alpha \cap D$.

Fix $x \in D$, $f(x) \in f(D) \implies \exists \alpha \in \Delta$ s.t. $f(x) \in O_\alpha \implies x \in f^{-1}(O_\alpha) = H_\alpha \cap D \subset H_\alpha$. Hence $\{H_\alpha\}_{\alpha \in \Delta}$ is an open cover for D.

Since D compact, $\exists \alpha_1, \alpha_2, \dots \alpha_k$ s.t. $D \subset H_{\alpha_1} \cup \dots \cup H_{\alpha_k}$.

Fix $y \in f(D), \exists x \in D$ s.t. $y = f(x)$. Since $x \in H_{\alpha_i}$ for some $i = 1, \dots k$, we have $x \in H_{\alpha_i} \cap D = f^{-1}(O_{\alpha_i}) \implies y = f(x) \in O_{\alpha_i}$. ✓

Corollary 4.19 (Extreme Value Theorem). Let D be compact, and $f : D \to \mathbb{R}$ be continuous. Then, f attains both its maximum and minimnum on D, i.e. $\exists x_1, x_2 \in D$ s.t. $f(x_1) \leq f(x) \leq f(x_2)$, $\forall x \in D$.

Proof. By theorem, $f(D)$ is compact.

Claim: Suppose $S \subset \mathbb{R}, S \neq \emptyset$ and S compact, then S contains its max and min.

Proof: since S bounded and nonempty $\sup S, \inf S$ exists. Since S closed, $\inf S, \sup S \in S$. ✓

Definition 4.20 (Uniform continuity)**.** Let $f : D \to \mathbb{R}$, we say f is uniformly continuous on D if $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $\forall x, y \in D$, $|x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

Remark. f can be continuous at any $c \in D$ but not uniformly continuous.

Example. $f(x) = \frac{1}{x}, D = (0, 1)$ is continuous but not uniformly continuous.

Example. $f(x) = \frac{1}{x}, D = (1, 2]$ is uniformly contiuous. Fix $\varepsilon > 0$, we seek $\delta > 0$ such that $|x - y| < \delta \implies |f(x) - f(y)| < \epsilon$.

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|x - y|}{x - y} \leq |x - y|$$

Hence we can pick $\delta = \varepsilon$.

Example. $f(x) = \sin x$ is uniform continuous on $\mathbb{R}$.

Example. $f(x) = x$ is uniform continuous on $\mathbb{R}$, $g(x) = x^2$ is not.

Remark. Boundedness is not required for uniform continuity. Consider $f(x) = x$.

Theorem 4.21. Let D be compact and $f : D \to \mathbb{R}$ be continuous on D. Then f is uniformly continuous on D.

Remark. On compact sets, continuity is necessary and sufficient for uniform continuity.

Proof. Fix $\varepsilon > 0$. $\forall x \in D$, since f continuous at x, $\exists \delta_x$ s.t.

$$|x - y| < \delta_x \implies |f(x) - f(y)| < \frac{\varepsilon}{2}$$

Fix $y_1, y_2 \in N(x, \delta_x)$. We want to say that $|f(y_1) - f(y_2)|$ is small enough.

Define $O_x = N\left(x, \frac{\delta_x}{2}\right)$. Then $\{O_x\}_{x \in D}$ is an open cover for D. By compactness, $\exists x_1, x_2, \ldots x_n \in D$ s.t. $D \subset O_{x_1} \cup \ldots O_{x_k}$. Define

$$\delta := \min\{\frac{\delta_{x_1}}{2}, \frac{\delta_{x_2}}{2}, \ldots, \frac{\delta_{x_k}}{2}, \}$$

Claim: $\forall x, y \in D, |x - y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

Fix x, y with $|x - y| < \delta$.

$$\begin{aligned} &|x - y| < \delta, \exists O_{x_i} \text{ s.t. } x \in O_{x_i} \\ \implies &|f(x) - f(x_i)| < \frac{\varepsilon}{2} \end{aligned}$$

Also

$$\begin{aligned} &|x - y| < \delta \leq \frac{\delta_{x_i}}{2} \\ \implies &|y - x_i| \leq |y - x| + |x - x_i| < \frac{\delta_{x_i}}{2} + \frac{\delta_{x_i}}{2} = \delta_{x_i} \\ \implies &y \in N(x_i, \delta_{x_i}) \\ \implies &|f(y) - f(x_i)| < \frac{\varepsilon}{2} \end{aligned}$$

Hence

$$|f(x) - f(y)| \leq |f(x) - f(x_i)| + |f(x_i) - f(y)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

✓

Definition 4.22. We say $\tilde{f} : E \to \mathbb{R}$ is an extension of a function $f : D \to \mathbb{R}$ if $D \subset E$ and $\tilde{f}(x) = f(x), \forall x \in D$.

Theorem 4.23. Let $f(a, b) \to \mathbb{R}$. TFAE

1. f is uniformly continuous on (a, b)
2. f is continuous on (a, b) and both $\lim_{x \to a^+} f(x)$ and $\lim_{x \to b^{-1}} f(x)$ exist in $\mathbb{R}$

3. f can be extended to a continuous function $\tilde{f}$ on $[a,b]$

Lemma 4.24 (Uniform continuity preserves Cauchyness)**.** Let $f: D \to \mathbb{R}$ be a uniformly continuous function on D, then for any Cauchy sequence $\{a_n\} \subset D$, $\{f(a_n)\}$ is also Cauchy.

Proof of 4.24. Let $\{a_n\}$ be Cauchy. WTS that $\{f(a_n)\}$ is Cauchy. $\forall \varepsilon > 0$, we seek $N > 0$ s.t. $|f(a_n) - f(a_m)| < \varepsilon, \forall n, m > N$.

Since f is uniform contiuous on D, $\exists \delta$ s.t. $|x-y| < \delta \implies |f(x) - f(y)| < \varepsilon$.

Since $\{a_n\}$ is Cauchy, $\exists N > 0$.s.t $\forall n, m > N$,

$$|a_n - a_m| < \delta \implies |f(a_n) - f(a_m)| < \varepsilon$$

✓

Proof of 4.23 .

1. (2) $\implies$ (3) **On homework, revisit**
2. (3) $\implies$ (2)
   - Fix any $c \in (a,b)$. WTS f continuous at c. Since c interior, $\exists \delta$ s.t. $N(c,\delta) \subset (a,b)$. Hence $f(x) = \tilde{f}(x)$ $\forall x \in N(c,\delta)$. $\tilde{f}$ continuous on $[a,b] \implies \tilde{f}$ continuous on $N(c,\delta) \implies f$ continuous on $N(c,\delta)$. Hence f continuous at c.
   - WTS $\lim_{x\to a^+} f(x)$ exists. Since $\tilde{f}$ continuous at a, $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} \tilde{f}(x) = f(a)$.
   - WTS $\lim_{x\to b^-} f(x)$ exists. Since $\tilde{f}$ continuous at b, $\lim_{x\to b^-} f(x) = \lim_{x\to b^-} \tilde{f}(x) = f(b)$.
3. (3) $\implies$ (1) **On homework, revisit**
4. (1) $\implies$ (2): it suffices to show that f uniformly continuous on (a,b) implies $\lim_{x\to a^+} f(x)$, $\lim_{x\to b^-} f(x)$ exist in $\mathbb{R}$. By symmetry, it suffices to show that $\lim_{x\to a^+} f(x)$ exists.

   ATC otherwise, i.e. $\lim_{x\to a^+} f(x)$ does not exist, then by theorem, $\exists \{a_n\} \subset (a,b)$ s.t. $a_n \to a^+$ and $\{f(a_n)\}$ diverges. However $\{a_n\}$ is Cauchy which means $\{f(a_n)\}$ must be Cauhcy. Hence a contradiction.

✓

4.4 Properties of continuous functions: connectedness and monotonicity

Theorem 4.25 (Intermediate Value Theorem)**.** Let f continuous, $f: [a,b] \to \mathbb{R}$, and y is a value in between $f(a), f(b)$. Then, $\exists c \in (a,b)$ s.t. $f(c) = y$.

Theorem 4.26 (Continuity preserves connectedness)**.** Let $f: D \to \mathbb{R}$ continuous and D is connected, then $f(D)$ is connected.

Corollary 4.27. Let I be a compact interval (i.e. closed and bounded $[a,b]$ or a singleton). If f is continuous on I, then $f(I)$ is a compact interval.

Proof of 4.26 $\implies$ IVT. Follows immediately by definition. Since $[a,b]$ is connected, $f([a,b])$ is connected. WLOG say $f(a) < f(b)$. For any $f(a) < y < f(b)$, $y \in f([a,b]) \implies \exists c \in [a,b]$ s.t. $f(c) = y$. ✓

Proof of 4.25. ATC $f(D)$ disconnected, $\exists A, B$ open s.t. $f(D) \subset A \cup B, A \cap f(D) \neq \emptyset$, $B \cap f(D) \neq \emptyset, A \cap B = \emptyset$. Since f continuous, and A open, $\exists$ open U s.t. $f^{-1}(A) = U \cap D$. Similarly, $\exists$ open V s.t. $f^{-1}(B) = V \cap D$. We then claim

1. $D \subset U \cup V$
2. $U \cap V \cap D = \emptyset$
3. $U \cap D \neq \emptyset, V \cap D \neq \emptyset$

Hence D is disconnected. ✓

Remark. Note that (2) above does not necessarily imply $U \cap V = \emptyset$. **Refer to notes on connectedness which shows that (2) is sufficient for disconnectedness.**

Definition 4.28 (Types of discontinuity)**.** Let $f: D \to \mathbb{R}, c \in D \cap D'$. Suppose f is discontinuous at c, then

1. f has a discontinuity of the **first type** if both $\lim_{x\to c^+} f(x)$ and $\lim_{x\to c^-} f(x)$ exists in $\mathbb{R}$
   - f either has a **jump** discontinuity or a **removable** discontinuity

2. Otherwise, f has a discontinuity of the **second type**

Definition 4.29 (Increasing, decreasing and monotone functions)**.** Let $f : (a, b) \to \mathbb{R}$, we say

- f is **increasing** on (a, b) if $a < x < y < b \implies f(x) \leq f(y)$.
- f is **decreasing** on (a, b) if $a < x < y < b \implies f(x) \geq f(y)$.
- f is **monotone** if it is either increasing or decreasing

Remark. We sometimes denote one-sided limits as

$$\lim_{x \to c^-} f(x) = f(c^-), \quad \lim_{x \to c^+} f(x) = f(c^+)$$

Proposition 4.30. If f is increasing on (a, b), then $\forall c \in (a, b)$, both $\lim_{x \to c^-} f(x)$ and $\lim_{x \to c^+} f(x)$ must exist, and

$$\lim_{x \to c^-} f(x) \leq f(c) \leq \lim_{x \to c^+} f(x)$$

In particular, f may not have discontinuities of the second kind on (a, b).

Proof. By assumption, $S := \{f(x) : a < x < c\}$ is bounded from above by $f(c)$. Also $S \neq \emptyset$. By CA, $\sup S = M$ exists, and $M \leq f(c)$.

Claim: $M = f(c-)$.

Fix $\varepsilon > 0$, seek δ s.t. $|c - x| < \delta \implies |f(x) - M| < \varepsilon$.

since $M = \sup S$, $\exists x_0 \in (a, c)$ s.t. $f(x_0) \in (M - \varepsilon, M]$. Pick $\delta = c - x_0$. $\forall x$ s.t. $x \in (c - \delta, c)$, we have

$$x_0 = c - \delta < x < c$$

Since f increasing,

$$f(x_0) \leq f(x) \leq f(M)$$

hence $|f(x) - M| < \varepsilon$. ✓

4.5 Limits involving infinity

Definition 4.31 (Limit at infinity)**.** Let $f : (c, \infty) \to \mathbb{R}$ for some $c \in \mathbb{R}$. We say the limit of f is $L \in \mathbb{R}$ at ∞ if $\forall \varepsilon > 0, \exists X > c$ s.t $x > X \implies |f(x) - L| < \varepsilon$, denoted

$$\lim_{x \to \infty} f(x) = L$$

Definition 4.32 (Diverges to infinity at a point)**.** Let f be defined on some neighborhood of $c \in \mathbb{R}$, say $N(c, \delta_0)$. We say f **diverges** to ∞ at c if $\forall M > 0, \exists \delta > 0$ s.t. $0 < |x - c| < \delta \implies f(x) > M$, denoted

$$\lim_{x \to c} f(x) = \infty$$

Definition 4.33 (Diverges to infinity at infinity)**.** Let $f : (c, \infty) \to \mathbb{R}$ for some $c \in \mathbb{R}$. We say f diverges to ∞ at ∞ if $\forall M > 0$, $\exists X$ s.t. $x > X \implies f(x) > M$, denoted

$$\lim_{x \to \infty} f(x) = \infty$$

5 Differentiation

5.1 Differentiability and derivatives

Definition 5.1 (Differentiability)**.** Let $I \subset \mathbb{R}$ be an interval, $f : I \to \mathbb{R}, c \in I$, f is said to be **differentiable** at c (**has a derivative** at c) if

$$\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

exists in $\mathbb{R}$. If so

$$f'(c) = \left.\frac{df}{dx}\right|_{x=c} = \lim_{x \to c} \frac{f(x) - f(c)}{x - c}$$

is called the **derivative** of f at c.

If f is differentiable at each point $S \subset I$, we say f is differentiable on S, and the function $f' : S \to \mathbb{R}$ is called the **derivative** of f on S.

Example. Fill in example from notes

Remark.

1. Unless c is on the end point of I, $f'(c)$ exists if and only if $f'_+(c)$ and $f'_-(c)$ exist and they agree.
2. A function can be continuous but not differentiable.

Theorem 5.2 (Differentiability implies continuity)**.** If $f : I \to \mathbb{R}$ is differentiable at $c \in I$, then f is continuous at c.

Proof. Since c is an accumulation point of I (**why??**), it suffices to show $f(c) = \lim_{x \to c} f(x)$.

For all $x \neq c$, one has

$$f(x) - f(c) = \frac{f(x) - f(c)}{x - c}(x - c)$$

Since $f'(c)$ exists, by limit theorem, we have

$$\lim_{x \to c} (f(x) - f(c)) = f'c \cdot \left(\lim_{x \to c}(x - c)\right) = f'(c) = 0$$

Hence

$$\lim_{x \to c} f(x) = \lim_{x \to c} f(c) = f(c)$$

✓

Remark. Differentiability implies continuity, but the converse is not true.

Example. Consider $f(x) = |x|$. f is continuous at $x = 0$ but not differentiable.

5.2 Basic Differentiation Rules and Formulas

Theorem 5.3. Suppose $f, g : I \to \mathbb{R}$ differentiable at $c \in I<$ then

1. $\forall k \in \mathbb{R}, (kf)'(c) = kf'(c)$
2. $(f + g)'(c) = f'(c) + g'(c)$
3. Product rule: $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$
4. Quotient rule: If $g(c) \neq 0$, then

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}$$

Proof of product rule.

$$\begin{aligned}
\lim_{x \to c} \frac{(fg)(x) - (fg)(c)}{x - c} &= \lim_{x \to c} \frac{f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)}{x - c} \\
&= \lim_{x \to c} f(x) \lim_{x \to c} \frac{g(x) - g(c)}{x - c} + = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \lim_{x \to c} g(c) \\
&= f(c)g'(c) + f'(c)g(c) \text{ since differentiability implies continuity}
\end{aligned}$$

✓

Theorem 5.4 (Chain Rule)**.** Let $f : I \to \mathbb{R}, g : J \to \mathbb{R}$ s.t. $f(I) \subset J$, where $I, J \subset \mathbb{R}$ are intervals. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and

$$(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$$

Proof of Quotient rule using Chain rule. Write $\frac{f}{g} = f \cdot \frac{1}{g}$ where $\frac{1}{g} = g \circ \frac{1}{x}$. **Claim**: If $x \neq 0$, then $\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$.
Proof:

$$\lim_{x \to c} \frac{\frac{1}{x} - \frac{1}{c}}{x - c} = \lim_{x \to c} \frac{-1}{xc} = -\frac{1}{c^2}$$

Hence by chain rule at c s.t. $g(c) \neq 0$, we have

$$\left(\frac{1}{g}\right)'(c) = -\frac{1}{(g(c))^2} g'(c)$$

By product rule,

$$\left(f \cdot \left(\frac{1}{g}\right)\right)'(c) = f'(c)\frac{1}{g(c)} + f(c) \cdot \frac{-g'(c)}{g^2(c)}$$

✓

Proof of Chain rule. Define a function $h : J \to \mathbb{R}$

$$h(y) = \begin{cases} \frac{g(y) - g(f(c))}{y - f(c)} & \text{if } y \neq f(c) \\ f'(f(c)) & \text{if } y = f(c) \end{cases}$$

This is an extension of $\frac{g(y)-g(f(c))}{y-f(c)}$ *to all* $y \in J$ *such that it is continuous at* $y = f(c)$.

By construction, we have $\forall y \in J$,

$$g(y) = g(f(c)) = h(y)(y - f(c))$$

Hence we have

$$\begin{aligned} \lim_{x \to c} \frac{g(f(x)) - g(f(c))}{x - c} &= \lim_{x \to c} \frac{h(f(x))(f(x) - f(c))}{x - c} \\ &= \lim_{x \to c} h(f(x)) \lim_{x \to c} \frac{f(x) - f(c)}{x - c} \end{aligned}$$

Where the second equality holds if both limits exist.

Claim: h is continuous at $f(c)$.
Proof: from construction of h,

$$\begin{aligned} \lim_{y \to f(c)} h(y) &= \lim_{y \to f(c)} \frac{g(y) - g(f(c))}{y - f(c)} \\ &= g'(f(c)) \text{ since } g \text{ differentiable at } f(c) \\ &= h(f(c)) \end{aligned}$$

Hence, we have f differentiable at $c \implies f$ continuous at $c \implies h(f(x))$ continuous at c, and

$$\lim_{x \to c} h(f(x)) = h(f(c)) = g'(f(c))$$

revisit this proof ✓

5.3 Mean Value Theorem

Theorem 5.5 (Mean value Theorem)**.** Let f be continuous on $[a, b]$, differentiable on (a, b) then there exists some $c \in (a, b)$ s.t.

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Theorem 5.6 (Rolle's Theorem)**.** Let f be continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then $\exists c \in (a, b)$ s.t. $f'(c) = 0$.

Corollary 5.7.

1. Let f be differentiable on (a,b) s.t. $f'(x)=0\ \forall x\in(a,b)$. Then f is constant on (a,b)
2. Let f,g be differentiable on (a,b) s.t. $f'(x)=g'(x),\forall x\in(a,b)$, then $f(x)=g(x)+C$ on (a,b) for some $C\in\mathbb{R}$

Proof. (1) $\implies$ (2): apply (1) to $f-g$. I.e. $h:=f-g$. Then $h=C$ for some C. $f=g+C$.

Proof of (1): Suppose f is not constant on (a,b), then $\exists x_1,x_2\in(a,b)$ s.t. $f(x_1)\neq f(x_2)$, say $x_1<x_2$. Then since f is continuous on $[x_1,x_2]$ and differentiable on (x_1,x_2), by MVT, $\exists c\in(x_1,x_2)\subset(a,b)$ s.t. $f'(c)=\frac{f(x_2)-f(x_1)}{x_2-x_1}\neq 0$. ✓

Corollary 5.8. Let f be differentiable on (a,b), then

1. $f'(x)\geq 0,\forall x\in(a,b)\iff f$ increasing on (a,b)
2. $f'(x)\leq 0,\forall x\in(a,b)\iff f$ decreasing on (a,b)

Remark. Note that

1. $f'>0$ on $(a,b)\implies f$ strictly increasing
2. $f'<0$ on $(a,b)\implies f$ strictly decreasing

But the converse is not true, e.g. $f=x^3$ is strictly increasing but $f'(0)=0$.

Proof of 5.8 .

- $\implies$: For any $a<x_1<x_2<b$, WTS $f(x_1)\leq f(x_2)$. By MVT on $[x_1,x_2]$, $\exists c\in(x_1,x_2)$ s.t.

$$0\leq f'(c)=\frac{f(x_2)-f(x_1)}{x_2-x_1}.$$

  Since $x_2-x_1>0$, we have $f(x_2)-f(x_1)\geq 0$
- $\impliedby$: Fix any $c\in(a,b)$, WTS $f'(c)\geq 0$. By definition,

$$f'(c)=\lim_{x\to c}\frac{f(x)-f(c)}{x-c}=\lim_{x\to c^+}\frac{f(x)-f(c)}{x-c}$$

  Sice $\forall x>c, f(x)\geq f(c)$, $f'(c)\geq 0$.

✓

Definition 5.9. Let $f:D\to\mathbb{R}$, f has a local maximum (minimum) at $p\in D$ if $\exists\delta>0$ s.t. $f(x)\leq f(p)$ (or $f(x)\geq f(p)$ for minimum), $\forall x\in D\cap N(p,\delta)$.

Remark.

1. If $p\in D$ is an isolated point, it is both the local max and local min.
2. f doesn't need ot be continuous at $p\in D$ to have local extrema, but f needs to be defined at p
3. If f constant on I, then it has local max or min everywhere on I.

Corollary 5.10. Let f be defined on $(a,b),p\in(a,b)$ then

1. If $\exists\delta>0$ s.t. f increasing on $(p-\delta,p]$ and decreasing on $[p,p+\delta)$, then f has a local max at p.
2. If f is differentiable on $N(p,\delta)\cap(a,b)$, for some $\delta>0$, then if $f'\geq 0$ on $(p-\delta,p)$ and $f'\leq 0$ on $(p,p+\delta)$, then f has local max at p.

Theorem 5.11 (Fermat Theorem)**.** Let $f:[a,b]\to\mathbb{R}$ and $p\in(a,b)$. Suppose f has a local max or local min at p and $f'(p)$ exists. Then $f'(p)=0$.

Remark. This theorem says that at an interior local extrema, the slope is zero. However, this doesn't apply to cases where

- extrema is at end point of domain
- extrema is at an isolated point
- extrema at point where f' undefined

Proof. Assume f has local max at $p\in(a,b)$, i.e. $\exists\delta>0$ s.t. $f(x)\leq f(p)$. $\forall x\in N(p,\delta)\cap[a,b]$. By choosing δ sufficiently small, we can have $N(p,\delta)\subset(a,b)$. Since f differentiable at p,

$$\lim_{x\to p}\frac{f(x)-f(p)}{x-p}=f'(p)$$

exists.

Since for all $p < x < p + \delta$, we have $x - p > 0, f(x) - f(p) \leq 0$, therefore we have

$$f'(p) = \lim_{x \to p^-} \frac{f(x) - f(p)}{x - p} \leq 0$$

By considering the left hand limit instead, we also have

$$f'(p) = \lim_{x \to p^-} \frac{f(x) - f(p)}{x - p} \geq 0$$

Hence $f'(p) = 0$. ✓

Remark. $f'(p) = 0 \not\Longrightarrow$ local extrema. e.g. $f = x^3$ at $x = 0$.

Proof of 5.5. *The idea is to first prove Rolle's using Fermat, and then use Rolle's to prove MVT.*

1. Step 1: WTS Rolle's

   Since f is continuous on $[a, b]$ which is compact, we know that f attains its max and min on $[a, b]$. i.e. $\exists x_1, x_2 \in [a, b]$ s.t. $\forall x \in [a, b]$

   $$f(x_1) \leq f(x) \leq f(x_2)$$

   Consider two cases

   - Case 1: both x_1, x_2 are end points of $[a, b]$. Then $f(a) = f(b) \implies f(x_1) = f(x_2) \implies f$ constant on $[a, b]$. Hence by definition, $f'(c) = 0$ for any $c \in [a, b]$.
   - Case 2: One of x_1, x_2 is not an endpoint of $[a, b]$. Say $x_1 \in (a, b)$. Then, f has a local min at x_1. By Fermat 5.11, $f'(x_1) = 0$. Take $c = x_1$.

2. Step 2: WTS Rolle's $\implies$ MVT

   Define

   $$h(x) = \frac{f(b) - f(a)}{b - a}(x - a) + f(a)$$
   $$g(x) = h(x) - f(x)$$

   Then observe that

   - f, h both continuous on $[a, b]$, hence g continuous on $[a, b]$
   - f', h', g' all exist on (a, b)
   - $h(a) = f(a), h(b) = f(b)$, hence $g(a) = g(b) = 0$

   Applying Rolle's to g, $\exists c \in (a, b)$ s.t. $g'(c) = 0$.

   $$g'(c) = f'(c) - h'(c) = f'(c) - \frac{f(b) - f(a)}{b - a}$$

   Hence

   $$f'(c) = \frac{f(b) - f(a)}{b - a}$$

✓

Proposition 5.12. Let $f : D \to \mathbb{R}$, if $f'(c) > 0$, then $\exists N(c, \delta)$ s.t.

- $f(x) < f(c), \forall x \in (c - \delta, c) \cap D$
- $f(c) < f(x), \forall x \in (c, c + \delta) \cap D$

Theorem 5.13 (IVT for derivatives)**.** Let f be differentiable on $[a, b]$ and k is strictly between $f'(a), f'(b)$. Then $\exists c \in (a, b)$ s.t. $f'(c) = k$.

Proof. WLOG, assume $f'(a) < k < f'(b)$. Let $g(x) \coloneqq f(x) - kx$, then g is differentiable on $[a, b]$ and $g'(x) = f'(x) - k$. WTS $\exists c \in (a, b)$ s.t. $g'(c) = 0$.

By Fermat, it suffices to show that g attains a local extrema somewhere.

Since g is continuous on $[a, b]$, which is compact, we know that the max and the min exist for the set $g([a, b])$, i.e. g attains global max and min on $[a, b]$.

We just need to exclude the case this global extrema is attained at the end points of $[a, b]$, in which case we cannot apply Fermat.

Claim: if g attains minimum at $c \in [a, b]$ then c is interior, i.e. $c \in (a, b)$.

Proof: since $f'(a) < k < f'(b)$, one has

$$g'(b) = f'(b) - k > 0 \implies \lim_{x \to b^-} \frac{g(x) - g(b)}{x - b} = g'(b) > 0$$

By definition of limit, $\exists \delta > 0$ s.t. $\forall x \in (b - \delta, b)$

$$\frac{g(x) - g(b)}{x - b} > 0$$

Since $x - b < 0$, we have $g(x) - g(b) < 0$. Hence g doesn't attain the minimum at b, i.e. $c \neq b$. Similarly, $c \neq a$.

Hence g attains local min at some $c \in (a, b)$, by Fermat, $g'(c) = 0$. ✓

Corollary 5.14. If f is differentiable on $[a, b]$ then f' may only have discontinuities of the second type on $[a, b]$.

Proof. **revisit this**

Suppose f' has a discountinuity of the first type at $c \in [a, b]$, say $c \in (a, b)$. Then $\lim_{x \to c^-} f'(x)$ $\lim_{x \to c^+} f'(x)$ both exist, but not both are equal to $f'(c)$. WLOG, say $\lim_{x \to c^-} f'(x) < f'(c)$. Then $\exists \epsilon, \delta$ s.t. $\forall x \in [c - \delta, c)$, $f'(x) < f'(c) - \varepsilon$.

Consider the interval $[c - \delta, c]$, and the number $k = f'(c) - \frac{\varepsilon}{2}$. f is differentiable on $[c - \delta, c]$, and k is strictly between $f'(c - \delta)$ and $f'(c)$. By theorem 3, $\exists x \in (c - \delta, c)$ s.t. $f'(x) = k$, which is a contradiction. ✓

Theorem 5.15 (Inverse Function Theorem)**.** Let f be differentiable on interval $I = (a, b)$ and $f'(x) \neq 0 < \forall x \in I$. Then

1. f is injective, i.e. f^{-1} can be defined as a function on $f(I)$
2. f^{-1} is differentiable on $f(I)$
3. The following is true

$$(f^{-1})'(y) = \frac{1}{f'(x)} = \frac{1}{f'\left(f^{-1}(y)\right)}, \forall y = f(x) \in f(I)$$

Example. **revisit these examples**

1. $g(x) = (x + 1)^{\frac{1}{3}}$, find $g'(7)$
2. Prove $(\arctan x)' = \frac{1}{1+x^2}$
3. Prove $(\arcsin x)' = \frac{1}{\sqrt{1-x^2}}$

Proof. Since $f' \neq 0$ on I, we must have either $f' > 0$ or $f' < 0$ on I

- Otherwise, if $f'(a) > 0, f'(b) < 0$, then by IVT for derivatives, $\exists c \in (a, b)$ s.t. $f'(c) = 0$

WLOG, assume $f' > 0$ on I, then f strictly increasing. Hence $x_1 \neq x_2 \implies f(x_1) \neq f(x_2) \implies f$ injective.

Fix any $y_0 \in f(I)$, WTS f' differentiable at y and

$$(f^{-1})'(y_0) = \frac{1}{f'\left(f^{-1}(y_0)\right)}$$

By injectivity, $\exists! x_0 \in I$ s.t. $f^{-1}(y_0) = x_0$. In fact $\forall y \in f(I), \exists! x_0 \in I$ s.t. $f^{-1}(y) = x$.

$$\begin{aligned}
\lim_{y \to y_0} \frac{f^{-1}(y) - f^{-1}(y_0)}{y - y_0} &= \lim_{y \to y_0} \frac{x - x_0}{f(x) - f(x_0)} \\
&= \lim_{y \to y_0} \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}} \text{ since } y \neq y_0 \implies x = f^{-1}(y) \neq f^{-1}(y_0) = x_0 \\
&= \lim_{x \to x_0} \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}} \text{by justification below} \\
&= \frac{1}{f'(x_0)} \\
&= \frac{1}{f'(f^{-1}(y_0))}
\end{aligned}$$

To justify

$$\lim_{y \to y_0} \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}} = \lim_{x \to x_0} \frac{1}{\frac{f(x) - f(x_0)}{x - x_0}}$$

, Let

$$G(x) = \begin{cases} \frac{1}{\frac{f(x)-f(x_0)}{x-x_0}} \text{ if } x \neq x_0, \\ \frac{1}{f'(x_0)} \text{if } x = x_0 \end{cases}$$

Since f differentiable, we know that G is continuous at x_0. **why??**
WTS

$$\lim_{y \to y_0} G\left(f^{-1}(y)\right) = G(x_0) = \frac{1}{f'(x_0)}$$

Claim: $\lim_{y \to y_0} f^{-1}(y) = x_0$
Proof: Sufficient for this is the continuity of f^{-1}. If f^{-1} continuous, we can pass the limit through a continuous function and say

$$\lim_{y \to y_0} G\left(f^{-1}(y)\right) = G\left(\lim_{y \to y_0} f^{-1}(y)\right) = G(x_0)$$

Hence it suffices to show that f^{-1} is continuous.

We know f continuous, and I is in interval. IVT implies $f(I)$ is also an interval. Furthermore, f is strictly monotone. By lemma below, f^{-1} is continuous. Hence we are done. ✓

Lemma 5.16. Let f be a strictly monotone and continuous function on interval I, then f^{-1} is continuous on $f(I)$.

In fact, f strictly increasing (decreasing) and continuous on interval I implies f^{-1} is also strictly increasing (decreasing) and continuous on $f^{-1}(I)$.

Proof. WLOG assume f strictly increasing. Then f^{-1} is well defined.

Claim: f^{-1} strictly increasing.

- Fix any $y_1 < y_2$. WTS $f^{-1}(y_1) < f^{-1}(y_2)$. Let $x_1 = f^{-1}(y_1)$, $x_2 = f^{-1}(y_2)$. Then obviously $x_1 < x_2$.

Next, WTS f^{-1} continuous on I. Fix any $y_0 \in f(I)$. $\exists\, !x_0 \in I$ s.t. $f(x_0) = y_0$. Fix $\varepsilon > 0$, we seek $\delta > 0$ s.t.

$$|y - y_0| < \delta \implies |f(y) - x_0| < \epsilon$$

For now, assume x_0 interior, $[x_0 - \varepsilon, x_0 + \varepsilon] \subset I$ **is this different from assuming x_0 interior?** Let $y_1 = f(x_0 - \varepsilon), y_2 = f(x_0 + \varepsilon)$. Define

$$\delta = \min\{|y_1 - y_0|, |y_2 - y_0|, \} > 0$$

Claim: $|y - y_0| < \delta \implies |f(y) - f(y_0)| < \varepsilon$.

- Fix any y s.t. $|y - y_0| < \delta$. Since f increasing, we have

$$y_1 = f(x_0 - \varepsilon) < f(x_0) = y_0 < f(x_0 + \varepsilon) = y_2$$

  Hence y is in the interval (y_1, y_2). Since $f(I)$ is an interval, and $y_1, y_2 \in I$, then $(y_1, y_2) \subset f(I)$. Hence f^{-1} is well defined on (y_1, y_2). How since f^{-1} increasing, $y_1 < y < y_2$ implies

$$x_0 - \varepsilon = f^{-1}(y_1) < f^{(y)} < f^{-1}(y_2) = x_0 + \varepsilon$$

✓

5.4 L'Hopital's Rule

Theorem 5.17 (L'Hopital's Rule, 0/0 case)**.** Let f, g be differentiable on $U = N^*(c, \delta)$ for some $\delta > 0$, s.t.

1. $\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = 0$
2. $g'(x) \neq 0, \forall x \in U$
3. $\lim_{x \to c} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{\pm\infty\}$

Then

$$\lim_{x \to c} \frac{f(x)}{g(x)} = L$$

Remark. The above conditions are sufficient for $g(x) \neq 0, \forall x \in U$. ATC otherwise, we can extend $g : N^*(x, \delta) \to \mathbb{R}$ to $\tilde{g} : N(x, \delta) \to \mathbb{R}$ by letting

$$\tilde{g}(c) = \lim_{x \to c} g(x) = 0$$

If $g(x) = 0$, then MVT on $[x, c]$ implies $\exists c_1 \in (x,)$ s.t.

$$g(c_1) = \frac{\tilde{g}(c) - \tilde{g}(x)}{c - x} = 0$$

which is a contradiction.

Lemma 5.18 (Cauchy Mean Value Theorem)**.** Let f, g be continuous on $[a, b]$, differentiable on (a, b), then $\exists c \in (a, b)$ s.t.

$$(f(b) - f(a)) \cdot g'(c) = (g(b) - g(a)) \cdot f'(c)$$

Remark.

1. $g(x) = x$ recovers MVT.
2. This is not a direct consequence of MVT

Proof. Define $h(x)$

$$h(x) = (f(b) - f(a))\, g(x) - (g(b) - g(a))\, f(x)$$

then h is continuous on $[a, b]$, differentiable on (a, b), and $h(a) = f(b)g(a) - g(b)f(a) = h(b)$. By Rolle's $\exists c \in (a, b)$ s.t. $h'(c) = 0$, i.e.

$$(f(b) - f(a)) \cdot g'(c) = (g(b) - g(a)) \cdot f'(c)$$

✓

Proof of 5.17 .

- Case $L \in \mathbb{R}$: Suffices to show

$$\lim_{x \to c^-} \frac{f(x)}{g(x)} = L$$

The idea is to

1. Use Cauchy MVT to write

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}$$

2. Taking the limit on both sides and show that the limit is equal to L.

Step 1: Fix any $x < c, x \in U$. Since we know that f, g differentiable and therefore continuous on $[x, c)$, and $\lim_{x \to c} f(x) = \lim_{x \to c} g(x) = 0$, we can define $\tilde{f}(c) = \tilde{g}(c) = 0$ and extend f, g to $\tilde{f}, \tilde{g}$ on $[x, c]$. We have $\tilde{f}, \tilde{g}$ differentiable on (x, c).

Applying Cauchy MVT, $\exists c_x \in (x, c)$ s.t.

$$\left(\tilde{f}(c) - \tilde{f}(x)\right) g'(c_x) = (\tilde{g}(c) - \tilde{g}(x))\, f'(c_x)$$

By assumption, $g' \neq 0$, which implies

- $g'(c_x) \neq 0$
- $\tilde{g}(c) - \tilde{g}(x) \neq 0$. Otherwise, by MVT, $\exists c_2 \in (x, c)$ s.t.

$$g'(c_2) = \frac{\tilde{g}(c) - \tilde{g}(x)}{c - x} = 0$$

Hence,

$$\left(\tilde{f}(c) - \tilde{f}(x)\right) g'(c_x) = (\tilde{g}(c) - \tilde{g}(x))\, f'(c_x) \implies \frac{\tilde{f}(c) - \tilde{f}(x)}{\tilde{g}(c) - \tilde{g}(x)} = \frac{f'(c_x)}{g'(c_x)}$$

Since $\tilde{f}(c) = \tilde{g}(c) = 0$, we have

$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)}$$

Step 2: by assumption, we have

$$\lim_{x \to c^-} \frac{f'(x)}{g'(x)} = L$$

hence $\forall \varepsilon, \exists \delta$ s.t.

$$x \in (c - \delta, c) \implies \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon$$

For any such x, since $x < c_x < c$, therefore $c - \delta < c_x < c$. Hence

$$\left| \frac{f'(c_x)}{g'(c_x)} - L \right| < \varepsilon$$

Therefore

$$\lim_{x \to c} \frac{f'(c_x)}{g'(c_x)} = L \implies \lim_{x \to c} \frac{f(x)}{g(x)} = L$$

- Case $L = \pm\infty$, WTS
$$\lim_{x\to c^+} \frac{f(x)}{g(x)} = \infty$$
As before, we get
$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)} \text{ for some } c_x \in (c, x)$$
By assumption,
$$\lim_{x\to c} \frac{f'(x)}{g'(x)} = \infty$$
implies $\forall M > 0, \exists \delta > 0$ s.t.
$$x \in (c, c+\delta) \implies \frac{f'(x)}{g'(x)} > M$$
Hence $\forall x \in (c, c+\delta)$, $c < c_x < c+\delta$, hence
$$\frac{f(x)}{g(x)} = \frac{f'(c_x)}{g'(c_x)} > M$$
which implieas
$$\lim_{x\to c^+} \frac{f(x)}{g(x)} = \infty$$

✓

Example. 1. $\lim_{x\to 0} \frac{e^x-1}{x}$ Check the conditions
- $\lim_{x\to 0} e^x - 1 = 0$, $\lim_{x\to 0} x = 0$
- $x' = 1 \neq 0$
- $f' = e^x, g' = 1$, $\lim_{x\to 0} \frac{f'}{g'} = \frac{e^x}{1} = 1$

Hence, L'Hopital gives
$$\lim_{x\to 0} \frac{e^x - 1}{x} = 1$$

2. $\lim_{x\to 0} \frac{\cos x}{x}$ Checking the conditions, $\lim_{x\to 0} \cos x = 1 \neq 0$, condition 1 fails.

Theorem 5.19 (L'Hopital's Rule for $x \to \infty$)**.** Let f, g be differentiable on (a, ∞) s.t.
1. $\lim_{x\to\infty} f(x) = \lim_{x\to\infty} g(x) = 0$
2. $g'(x) \neq 0, \forall x \in (a, \infty)$
3. $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{\pm\infty\}$

Then,
$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L$$

Remark. Again $g(x) \neq 0$ is implicitly given.
- if a single point $g(x_0) = 0$, and $g(x) \neq 0, \forall x \neq x_0$, then we can consider the theorem on (x_0, ∞).
- if a sequence $\{a_n\}$ s.t. $f(a_n) = 0$, then we can find a contradiction with MVT **do we need sequence to find contradiction?? or is it just two points that's needed??**

Example. **add in example**

Proof. *The idea is to use change of variables and the first form of L'Hopital.*
Let $x = \frac{1}{t}$, then $x \to \infty \iff t \to 0^+$. Define $F(t) = f\left(\frac{1}{t}\right)$, $G(t) = g\left(\frac{1}{t}\right)$.
We first show that satisfying the conditions of 5.19 implies the conditions of 5.17.
1. By assumption, we have $\lim_{x\to\infty} f(x) = \lim_{g\to\infty} g(x) = 0$
$$\lim_{x\to\infty} f(x) = \lim_{g\to\infty} g(x) = 0$$
$$\implies \lim_{t\to 0^+} f\left(\frac{1}{t}\right) = \lim_{g\to 0^+} g\left(\frac{1}{t}\right) = 0$$
$$\implies \lim_{t\to 0^+} F(t) = \lim_{g\to 0^+} G(t) = 0$$

2. By assumption, $g'(x) \neq 0$. Consider $G'(t)$

$$G'(t) = g'\left(\frac{1}{t}\right) \cdot \frac{-1}{t^2}$$
$$\implies G'(t) \neq 0$$

3. By assumption, $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = L$. Then, by change of variables, we have

$$\lim_{t\to 0^+} \frac{f'\left(\frac{1}{t}\right)}{g'\left(\frac{1}{t}\right)} = \lim_{x\to\infty} \frac{f'(x)}{g'x} = L$$

By Chain Rule,

$$F'(t) = f'\left(\frac{1}{t}\right) \cdot \left(-\frac{1}{t^2}\right)$$
$$G'(t) = g'\left(\frac{1}{t}\right) \cdot \left(-\frac{1}{t^2}\right)$$

Hence

$$\lim_{x\to 0^+} \frac{F'(t)}{G'(t)} = \lim_{x\to 0^+} \frac{f'\left(\frac{1}{t}\right)}{g'\left(\frac{1}{t}\right)} = L$$

Having verified all 3 conditions, apply 5.17 to F, G, and we get

$$\lim_{t\to 0^+} \frac{F(t)}{G(t)} = \lim_{t\to 0^+} \frac{F'(t)}{G'(t)} = L$$

Applying change of variables again, we get

$$\lim_{x\to [infty]} \frac{f(x)}{g(x)} = \lim_{t\to 0^+} \frac{F(t)}{G(t)} = L$$

✓

Remark. The first two L'Hopital's Rule theorems (5.17, 5.19) theoretically cover all possible cases of limits of indeterminate forms. However, for convenience, we extend L'Hopital's to $\frac{\infty}{\infty}$ forms directly.

Theorem 5.20 (L'Hopital's Rule for $\frac{\cdot}{\infty}$)**.** Let f, g be differentiable on $U = N^*(c, \delta)$ for some $\delta > 0$ s.t.

1. $\lim_{x\to c} g(x) = \pm\infty$
2. $g'(x) \neq 0, \forall x \in U$
3. $\lim_{x\to c} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{\pm\infty\}$

Then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = L$$

Remark. Note that 5.20 does not require $\lim_{x\to c} f(x) = \pm\infty$. This makes the theorem stronger, but means we cannot prove using change of variables and 5.17.

Theorem 5.21 (L'Hopital's Rule for $\frac{\cdot}{\infty}, x \to \infty$)**.** Let f, g be differentiable on (a, ∞) s.t.

1. $\lim_{x\to\infty} g(x) = \pm\infty$
2. $g'(x) \neq 0, \forall x \in (a, \infty)$
3. $\lim_{x\to\infty} \frac{f'(x)}{g'(x)} = L \in \mathbb{R} \cup \{\pm\infty\}$

then,

$$\lim_{x\to\infty} \frac{f(x)}{g(x)} = L$$

Remark. 5.21 follows from 5.20 by change of variables.

5.5 Taylor's Theorem

Theorem 5.22 (Taylor's Theorem)**.** Let $f^{(i)}$ be continuous on $[a,b]$, differentiable on (a,b), $\forall 0 \le i \le n$. Let $x_0 \in [a,b]$. Then $\forall x \in [a,b]$ with $x \neq x_0$, $\exists c$ between x and x_0 s.t.

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f^{(2)}(x_0)}{2!}(x-x_0)^2 + \ldots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + \frac{f^{(n+1)}(c)}{n!}(x-x_0)^n$$

why is it always continuous on closed interval and differentiable on open interval??

Remarks.
- Since the theorem requires $f^{(n)}$ differentiable, the RHS ($f^{(n+1)}$) is well defined.
- Since c depends on x, the RHS is not in a polynomial form.
- If $n = 0$, we recover MVT.

Proof. Fix $x \in [a,b], x \neq x_0$. Let M be the unique solution of

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f^{(2)}(x_0)}{2!}(x-x_0)^2 + \ldots + \frac{f^{(n)}(x_0)}{n!}(x-x_0)^n + M(x-x_0)^n$$

. WTS $M = \frac{f^{(n+1)}(c)}{(n+1)!}$ for some c between x, x_0. Define

$$F(t) = f(t) + f'(t)(x-t) + \ldots + \frac{f^{(n)}(t)}{n!}(x-t)^n + M(x-t)^{n+1}$$

Then F is continuous on $[a,b]$, differentiable on (a,b). Observe that

$$F(x) = f(x), F(x_0) = f(x)$$

By Rolle's theorem, $\exists c$ between x, x_0 s.t. $F'(c) = 0$. Differentiating F, we get

$$\begin{aligned} F'(c) =& f'(c) + \left[-f'(c) + f^{(2)}(c)(x-c)\right] \\ &+ \left[-2\frac{f^{(x)}(c)}{2!}(x-c) + \frac{f^{(3)}(c)}{2!}(x-c)^2\right] \\ &+ \left[-3\frac{f^{(3)}(c)}{3!}(x-c)^2 + \frac{f^{(4)}(c)}{3!}(x-c)^3\right] \\ &+ \ldots \\ &+ \left[-n\frac{f^{(n)}(c)}{n!}(x-c)^n + \frac{f^{(n+1)}(c)}{n!}(x-c)^n\right] \\ &+ (-M)(n+1)(x-t)^n \end{aligned}$$

Setting this to 0, we get

$$M = \frac{f^{(n+1)}(c)}{(n+1)!}$$

✓

Definition 5.23 (Taylor's Polynomials)**.** Let $f^{(i)}$ be continuous on $[a,b]$, differentiable on (a,b), $\forall 0 \le i \le n>$ Let $x_0 \in [a,b]$. Fix any $x \in [a,b]$, $x \neq x_0$, then the **Taylor's Polynomials** for f at x_0 are

$p_0(x) = f(x_0)$
$p_1(x) = f(x_0) + f'(x_0)(x-x_0)$
$p_2(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f^{(2)}(x)}{2!}(x-x_0)^2$
$\vdots$
$p_2(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{f^{(2)}(x)}{2!}(x-x_0)^2 + \ldots \frac{f^{(n)}(x)}{n!}(x-x_0)^n$

Remark. What is the relationship between f and $\{p_n\}$
- If f is a polynomial, then $\exists N$ s.t. $p_n = f, \forall n \ge N$
- For general f, $p_n(x)$ is an approximation

- of f with an upper bound on the magnitude of error

$$\frac{\max \left|f^{(n+1)}\right|}{(n+1)!} |x - x_0|^{n+1}$$

6 Infinite Series

6.1 Basic properties of series

Definition 6.1 (Infinite Series)**.** Given a sequence $\{a_n\} \subset \mathbb{R}$. An infinite series is defined as

$$\sum_{k=1}^{\infty} a_k$$

Definition 6.2 (Partial sum)**.** Given a sequence $\{a_n\} \subset \mathbb{R}$. For all $n \geq 1$, the n**-th partial sum** is defined as

$$S_n \coloneqq \sum_{k=1}^{n} a_k$$

The set $\{S_n\}_{n\in\mathbb{N}}$ is called the **partial sums** of the series $\sum_{n=1}^{\infty} a_n$.

Definition 6.3 (Convergence of a series)**.** Given a sequence $\{a_n\} \subset \mathbb{R}$, and partial sums $\{S_n\}_n$,

- we say the series $\sum_{k=1}^{\infty} a_k$ converges to $S \in \mathbb{R}$ if $\{S_n\}$ converges to $S \in \mathbb{R}$, this is denoted

$$\sum_{n=1}^{\infty} a_n = S$$

- we say the series diverges if $\{S_n\}$ diverges.

Example (Geometric Series)**.** Fix $a \in \mathbb{R}$, its geometric series is

$$\sum_{n=1}^{\infty} a^{n-1} = 1 + a + a^2 + \dots$$

This converges to $\frac{1}{1-a}$ if $|a| < 1$, and diverges otherwise.

Proof: Recall

$$\sum_{k=1}^{n} a^{k-1} = 1 + a + a^2 + \dots a^n = \frac{1-a^n}{1-a}$$

Consider 3 cases

- $|a| < 1$: WTS $\lim_{n\to\infty} a^n = 0$ Fix $\varepsilon > 0$, take $N = \frac{\ln \varepsilon}{\ln |a|}$. Then $\forall n > N$

$$n > \frac{\ln \varepsilon}{\ln |a|}$$
$$\implies n \ln |a| < \ln \varepsilon \implies |a|^n < \varepsilon \implies |a^n| < \varepsilon$$

  and we are done. Hence

$$\sum_{k=1}^{\infty} a^{k-1} = \lim_{n\to\infty} \frac{1-a^n}{1-a} = \frac{1}{1-a}$$

- $|a| = 1$: $S_n = n$, diverges.
- $|a| > 1$: $\lim_{n\to\infty} a^n$ diverges.

Theorem 6.4 (Series converges iff sequence of partial sums is Cauchy)**.** $\sum_{n=1}^{\infty} a_n$ converges if and only if $\forall \varepsilon > 0, \exists N > 0$ s.t. $m \geq n > N \implies \left| \sum_{k=n}^{m} a_k \right| < \varepsilon$.

Proof. By definition and Cauchy Convergence Criterion,

$$\begin{aligned}
&\sum_{n=1}^{\infty} a_n \text{ converges} \\
\iff &\{S_n\} \text{ converges} \\
\iff &\{S_n\} \text{ Cauchy} \\
\iff &\forall \varepsilon > 0, \exists N > 0 s.t. \forall m > n > N, |S_n - S_m| = \left|\sum_{k=n+1}^{m} a_k\right| < \varepsilon \\
\iff &\forall \varepsilon > 0, \exists N > 0 s.t. \forall m \geq n > N_1, \left|\sum_{k=n}^{m} a_k\right| < \varepsilon
\end{aligned}$$

The last equivalence (more specifically, its forward direction, requires a relabling, **revisit this**). ✓

Corollary 6.5 (Convergence of series implies convergence of sequence to 0)**.** If $\sum_{n=1}^{m} a_n$ converges then $\lim_{n\to\infty} a_n = 0$.

Proof. By theorem, $\forall \varepsilon > 0, \exists N > 0$ s.t. $\forall m \geq n > N$

$$\left|\sum_{k=n}^{m} a_k\right| < \varepsilon$$

Take $m = n$, then $|a_n| < \varepsilon$. Hence $\lim_{n\to\infty} a_n = 0$. ✓

Remark.
- The converse is not true. Convergence of a sequence to 0 does not imply convergence of series. For example, harmonic series.
- The corollary is useful for showing a series diverges, by using its contrapositive.

Definition 6.6 (Absolute and conditional convergence)**.**

1. We say $\sum_{n=1}^{\infty} a_n$ **converges absolutely** if $\sum_{n=1}^{\infty} |a_n|$ converges.
2. If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} |a_n|$ diverges, then we say $\sum_{n=1}^{\infty} a_n$ **converges conditionally**.

Corollary 6.7 (Absolute convergence implies convergence)**.** If $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_n$ converges.

Proof. Suffices to verify that the sequence of partial sums is Cauchy.

Fix $\varepsilon > 0$, since $\sum |a_n|$ converges, $\exists N > 0$ s.t.

$$\begin{aligned}
m \geq n > N &\implies \left|\sum_{k=n}^{m} |a_k|\right| < \varepsilon \\
&\implies \sum_{k=n}^{m} |a_k| < \varepsilon
\end{aligned}$$

By triangle inequality,

$$\left|\sum_{k=n}^{m} a_k\right| \leq \sum_{k=n}^{m} |a_k| < \varepsilon$$

✓

Proposition 6.8. Let $\sum_{n=1}^{\infty} a_n$ and $\sum_{n=1}^{\infty} b_n$ converge, and $\alpha, \beta \in \mathbb{R}$, then

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n) = \alpha \sum_{n=1}^{\infty} a_n + \beta \sum_{n=1}^{\infty} b_n$$

Proof. Let $A_n = \sum_{k=1}^{n} a_k$, $B_n = \sum_{k=1}^{n} b_k$. By assumption ✓

Remark. It is not generally true that

$$\sum_{n=1}^{\infty}(a_n b_n) = \left(\sum_{n=1}^{\infty} a_n\right)\left(\sum_{n=1}^{\infty} b_n\right)$$

6.2 Nonnegative series

Definition 6.9 (Nonnegative series)**.** If $a_n \geq 0, \forall n \in \mathbb{N}$, then $\sum_{n=1}^{\infty} a_n$ is called a **nonnegative series**.

Theorem 6.10 (Nonnegative series, theorem 1)**.** Let $\sum a_n$ be a nonnegative series, then

1. $\sum a_n$ converges $\iff$ its partial sums $\{S_n\}$ are bounded
2. If $\sum a_n$ diverges, then

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = \infty$$

Proof. $\{S_n\}$ is a nonnegative increasing sequence.

1. By MCT, since $\{S_n\}$ monotone, it converges if and only if it is bounded.
2. $\sum a_n$ diverges, then by (1), $\{S_n\}$ unbounded from above. Hence $\forall M > 0, \exists N$ s.t. $S_N > M$, and $\forall n > N, S_n \geq S_N > M$. Hence $\lim_{n\to\infty} S_n = \infty$.

✓

Theorem 6.11 (Nonnegative Series, Theorem 2)**.** Given a series $\sum a_n$ and let $\sum b_k$ be a series obtained from a regrouping of $\sum a_n$ (obtained by adding paranthesis), then

1. $\sum a_n$ converges $\implies$ $\sum b_k$ converges and $\sum a_n = \sum b_k$
2. If $\sum a_n$ is nonnegative series, then $\sum a_n = A \iff \sum b_k = A$.

Remark. When a_n nonnegative, convergence of series and convergence of any regrouping is equivalent.

For general a_n, we only have the forward direction. The converse (convergence of regrouping implying convergence of original series) is NOT true.

Proof. Let S_n and t_n denote the n-th partial sum of $\sum a_n$ and $\sum b_k$ respectively. Then $\{t_n\}$ is a subsequence of $\{S_n\}$.

For the forward direction, which holds for any convergent $\sum a_n$

$$\begin{aligned}\sum a_n \text{ converges} &\implies \{S_n\} \text{ converges} \\ &\implies \{t_n\} \text{ converges} \\ &\implies \sum b_k \text{ converges}\end{aligned}$$

For the reverse direction, $\sum a_n$ nonnegative, $\sum b_k$ converges. Since S_n is increasing and has a convergent subsequence t_n, therefore S_n must converge to the same limit.

- t_n converges implies t_n bounded.
- S_n monotone, and has a bounded subsequence t_n, hence S_n is bounded.
- By MCT, S_n converges.

✓

Corollary 6.12. Let $\{S_n\}$ be the sequence of partial sums of nonnegative series $\sum a_n$. If $\{S_n\}$ has a bounded subsequence (or equivalently convergent) then $\sum a_n$ converges.

6.3 Convergence Rules for Nonnegative series

Theorem 6.13 (Comparison test)**.** Let $\sum a_n, \sum b_n$ be two nonnegative series. If $\exists N > 0$ s.t. $\forall n > N, a_n \leq b_n$, then

1. $\sum b_n$ converges $\implies$ $\sum a_n$ converges.
2. $\sum a_n$ diverges $\implies$ $\sum b_n$ diverges.

Proof. WLOG, assume $a_n \leq b_n, \forall n \in \mathbb{N}$. **Check why we can do this.**
Proof of (1): Let S_n^a, S_n^b denote the n-th partial sums of $\sum a_n, \sum b_n$.

$$\begin{aligned} &\sum b_n \text{ converges} \\ \implies &S_n^b \text{ bounded} \\ \implies &\exists M > 0 \text{ s.t. } |S_n^b| \leq M, \forall n \in \mathbb{N} \\ \implies &|S_n^a| \leq \left|S_n^b\right| \leq M, \forall n \in \mathbb{N} \\ \implies &S_n^a \text{ bounded} \\ \implies &\sum a_n \text{ converges} \end{aligned}$$

Proof of (2):

$$\begin{aligned} &\sum a_n \text{ diverges} \\ \implies &S_n^a \text{ unbounded} \\ \implies &S_n^b \text{ unbounded} \\ \implies &\sum b_n \text{ diverges} \end{aligned}$$

✓

Example (p-series)**.** $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converges if $p > 1$, diverges if $p \leq 1$.

Proof:

Step 1: $p > 1$, suffices to show a regrouping converges. Consider

$$S_{2^k-1} = \frac{1}{1} + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{(2^k-1)p}$$

Let $b_k = S_{2^k-1} - S_{2^{k-1}-1}, \forall k \geq 2$, and $b_1 = a_1$. i.e.

$$\begin{aligned} b_1 &= a_1 \\ b_2 &= S_3 - S_1 = a_2 + a_3 \\ b_3 &= S_7 - S_3 = a_4 + a_5 + a_6 + a_7 \end{aligned}$$

WTS $\sum b_k$ converges. We use comparison test with geometric series.

$$\begin{aligned} b_k &= S_{2^k-1} - S_{2^{k-1}-1} \\ &= \sum_{n=2^{k-1}}^{2^k} \frac{1}{n^p} \\ &\leq \frac{1}{(2^{k-1})^p}(2^k - 2^{k-1}) \\ &= \frac{2^{k-1}}{(2^{k-1})^p} \\ &= (2^{k-1})^{1-p} \\ &= \left(\frac{1}{2^{p-1}}\right)^{k-1} \end{aligned}$$

From geometric series, we know that

$$\sum_{k=1}^{\infty} \left(\frac{1}{2^{P-1}}\right)^{k-1}$$

converges, hence $\sum b_k$ converges.

Step 2: $p \leq 1$: since $\frac{1}{n^p} \geq \frac{1}{n}$, by comparison test, $\sum \frac{1}{n^p}$ diverges.

Theorem 6.14 (Ratio test)**.** Let $\sum a_n$ be a nonnegative series, $a_n \neq 0$. Then

1. If $\limsup_{n\to\infty} \frac{a_{n+1}}{a_n} < 1$, then $\sum a_n$ converges.
2. If $\liminf_{n\to\infty} \frac{a_{n+1}}{a_n} > 1$, then $\sum a_n$ diverges.

Remark. If the limsup or liminf is 1, then it is inconclusive.

Example. $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges, by ratio test.

$$\frac{a_{n+1}}{a_n} = \frac{n+1}{2^{n+1}}\frac{2^n}{n} = \frac{1}{2}\frac{n+1}{n} \to \frac{1}{2}$$

Proof.

1. Let $r \in (\limsup_{n\to\infty} \frac{a_{n+1}}{a_n}, 1)$, then $\exists N > 0$ s.t. $\forall n > N$, $\frac{a_{n+1}}{a_n} < r$. Hence

$$\begin{aligned} a_{n+1} &= \frac{a_{n+1}}{a_n}\frac{a_n}{a_{n-1}}\frac{a_{n-1}}{a_{n-2}} \cdots \frac{a_{N+2}}{a_{N+1}} a_{N+1} \\ &\leq r^{n-N} a_{N+1} \\ &= r^n \frac{a_{N+1}}{r^N} \\ &= cr^n \text{ for some constant } c \end{aligned}$$

   Since $r < 1 \implies \sum cr^n$ converges, by comparison test, $\sum a_n$ converges **revisit this**

2. **fill in mirrored proof**

✓

Theorem 6.15 (Root test)**.** Let $\sum a_n$ be a nonnegative series, then

1. If $\limsup_{n\to\infty} (a_n)^{\frac{1}{n}} < 1$ then $\sum a_n$ converges.
2. If $\limsup_{n\to\infty} (a_n)^{\frac{1}{n}} > 1$ then $\sum a_n$ diverges.

Proof.

1. Let $r \in \left(\limsup_{n\to\infty} a_n^{\frac{1}{n}}, 1\right)$ then $\exists N$ s.t $\forall n > N$, $a_n^{1/n} < r \implies a_n < r^n$. Since $r < 1$, $\sum r^n$ converges. Hence

by comparison test, $\sum a_n$ converges **revisit this**

2. **fill in mirrored proof**

✓

Example (Example where ratio test fails but root test works)**.** Consider the series

$$\frac{1}{2}+\frac{1}{3}+\frac{1}{2^2}+\frac{1}{3^2}+\frac{1}{2^3}+\frac{1}{3^3}+\dots$$

that is, $a_{2k-1}=\frac{1}{2^k}$ and $a_{2k}=\frac{1}{3^k}$ for $k\geq 1$.

Ratio test: the ratios alternate between two behaviours.

$$\frac{a_{2k}}{a_{2k-1}}=\frac{1/3^k}{1/2^k}=\left(\frac{2}{3}\right)^k\to 0$$
$$\frac{a_{2k+1}}{a_{2k}}=\frac{1/2^{k+1}}{1/3^k}=\frac{1}{2}\left(\frac{3}{2}\right)^k\to\infty$$

Hence $\liminf_{n\to\infty}\frac{a_{n+1}}{a_n}=0$ and $\limsup_{n\to\infty}\frac{a_{n+1}}{a_n}=\infty$. Neither $\limsup<1$ nor $\liminf>1$ holds, so the ratio test does not apply.

Root test: $\limsup$ is the largest subsequential limit, so we compute the limit of $\sqrt[n]{a_n}$ along the odd and the even terms separately.

- $n=2k-1$ (odd), $a_n=\frac{1}{2^k}$:

$$\sqrt[n]{a_n}=\left(\frac{1}{2^k}\right)^{\frac{1}{2k-1}}=2^{-\frac{k}{2k-1}}=2^{-\frac{1}{2-1/k}}\xrightarrow{k\to\infty}2^{-1/2}=\frac{1}{\sqrt{2}}$$

- $n=2k$ (even), $a_n=\frac{1}{3^k}$:

$$\sqrt[n]{a_n}=\left(\frac{1}{3^k}\right)^{\frac{1}{2k}}=3^{-1/2}=\frac{1}{\sqrt{3}}$$

These are the only subsequential limits, so

$$\limsup_{n\to\infty}\sqrt[n]{a_n}=\max\left\{\frac{1}{\sqrt{2}},\frac{1}{\sqrt{3}}\right\}=\frac{1}{\sqrt{2}}<1$$

By the root test, the series converges.

6.4 Discussion of e

Definition 6.16 (e)**.** e is defined as the sum of the infinite series

$$e:=\sum_{n=0}^{\infty}\frac{1}{n!}=1+\frac{1}{1!}+\frac{1}{2!}+\dots$$

Add in discussion of e.

6.5 Alternating Series

Theorem 6.17. Let $\{a_n\}$ be a monotone sequence such that $\lim_{n\to\infty}a_n=0$. Then the series

$$\sum_{n=1}^{\infty}(-1)^n a_n$$

converges.

Example. $\sum_{n=1}^{\infty}(-1)^n\frac{1}{n}=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}$ converges.

Theorem 6.18. Let series $\sum a_n$ satisfy:

1. $\lim_{n\to\infty}a_n=0$

2. $\exists N > 0$ and a convergent regrouping $\sum b_k$ s.t. b_k is the sum of $\leq N$ terms of $\{a_n\}, \forall k \in \mathbb{N}$

Then $\sum a_n$ converges.

Proof of 6.18. Let S_n be the n-th partial sum of $\sum a_n$, and let S_{n_k} be its subsequence that represents the k-th partial sum of $\sum b_k$. Since $\sum b_k$ converges, $\exists L \in \mathbb{R}$ s.t.

$$\lim_{k\to\infty} S_{n_k} = L$$

Claim: $\lim_{n\to\infty} S_n = L$

Fix any $n \in \mathbb{N}$, then $\exists k \in \mathbb{N}$ s.t. $n_k \leq n \leq n_{k+1}$, hence

$$|S_n - S_{n-k}| \leq \sum_{j=1}^{N} |a_{n_k+j}|$$

Fix any $\varepsilon > 0$, seek N_0 s.t. $|S_n - L| < \varepsilon, \forall n > N_0$.

Let $K_1 > 0$ s.t. $\forall k \geq K_1, |S_{n_k} - L| < \frac{\varepsilon}{2}$. Let $N_1 > 0$ s.t. $\forall n > N_0, |a_n| < \frac{\varepsilon}{2N}$.

Take $N_0 = \max\{n_{k+1}, N_1 + N\}$ then $\forall n > N_0$. the unique k s.t. $n_k \leq n \leq n_{k+1}$ must satisfy $k \geq K_1$. Hence $|S_{n_k} - L| < \frac{\varepsilon}{2}$. Also, $n > N_0 \geq N_1 + N \implies, n_k > N_1$, hence

$$|a_{n_k+j}| < \frac{\varepsilon}{2N}$$

By triangle inequality,

$$\begin{aligned} |S_n - L| &\leq |S_n - S_{n_k}| + |S_{n_k} - L| \\ &\leq \sum_{j=1}^{N} |a_{n_k} + j| + |S_{n_k} - L| \\ &\leq N \cdot \frac{\varepsilon}{2N} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

✓

Proof of 6.17. WLOG, assume $\{a_n\}$ decreasing, then since $\lim_{n\to\infty} a_n = 0$, we know that $a_n \geq 0, \forall n$. We want to find a regrouping b_k and apply 6.17.

The sequence we are given is

$$\sum_{n=1}^{\infty} (-1)^n a_n = -a_1 + a_2 - a_3 + a_4 + \dots$$

For convenience, we consider

$$\sum_{n=1}^{\infty} (-1)^{n-1} a_n = a_1 - a - 2 + a_3 - a_4 + \dots$$

Since a_n decreasing, we have

$$\begin{aligned} -\sum_{n=1}^{\infty} (-1)^{n-1} a_n &= a_1 - a - 2 + a_3 - a_4 + \dots \\ &= (a_1 - a - 2) + (a_3 - a_4) + \dots \\ &= b_1 + b_2 + \dots \end{aligned}$$

where $k = 2n$, and

$$b_k = (-1)^{(2k-1)-1} a_{2k-1} + (-1)^{2k-1} a_{2k}$$

Then we verify the two conditions of 6.18

- $\lim_{n\to\infty} a_n = 0$, given
- there exists a convergent regrouping $\sum_k b_k$ where each each regrouping is the sum of two terms of a_n
  - To show $\sum b_k$ convergent, since b_k nonnegative, suffices to show its partial sums are bounded. Consider

S_k, the k-th partial sum of b_k

$$\begin{aligned} S_k &= \sum_{i=1}^{k} b_i \\ &= (a_1 - a_2) + (a_3 - a_4) + \ldots + (a_{2k-1}, a_{2k}) \\ &= a_1 - (a_2 - a_3) - (a_4 - a_5) \ldots - (a_{2k-2} - a_{2k-1}) - a_{2k} \\ &\leq a_1 \end{aligned}$$

Hence we apply 6.18 and say that $\sum(-1)^{n-1}a_n$ converges.

✓

Theorem 6.19 (Dirichlet Convergence Test)**.** Let

- $\sum_{n=1}^{\infty} a_n$ be a series with uniformly bounded partial sums
- $\{b_n\}$ be a monotone sequence s.t. $b_n \to 0$

Then $\sum_{n=1}^{\infty} a_n b_n$ converges and

$$\left|\sum_{n=1}^{\infty} a_n b_n\right| \leq A\,|b_1|$$

6.6 Products and rearrangement of series

Definition 6.20 (Rearrangement of series)**.** Let $\sum a_n$ be a series.

1. $\sum a'_n$ is a **rearrangement** of $\sum a_n$ if there is a bijection $f : \mathbb{N} \to \mathbb{N}$ s.t. $a'_n = a_{f(n)}, \forall n \in \mathbb{N}$.
2. $\sum a_{f(n)}$ is called a **bounded rearrangement** of $\sum a_n$ if $\exists M > 0$ s.t. $|f(n) - n| \leq M, \forall n \in \mathbb{N}$.

Theorem 6.21 (Series converges iff every bounded rearrangement converges)**.** Let $\sum a_{f(n)}$ be a bounded rearrangement of $\sum a_n$. Then

$$\sum a_n \text{ converges} \iff \sum a_{f(n)} \text{ converges}$$

and if they converge then they converge to the same limit

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{f(n)}$$

Proof. Suffices to show

- $\sum a_n \implies \sum a_{f(n)}$
- $\sum a_n = \sum a_{f(n)}$

Fix convergent $\sum a_n$ and bounded rearrangement f, i.e. $|f(n) - n| \leq M$.

Since $\sum a_n$ converges, $a_n \to 0$, we get

$$\lim_{n\to\infty} \sum_{-M}^{M} |a_{n+j}| = 0$$

The proof is to fix ε, and pick N s.t. $|a_n| < \frac{\varepsilon}{2M+1}$.

Let S_n, S'_n denote the n-th partial sum of $\sum a_n, \sum a_{f(n)}$, then

$$|S_n - S'_n| \leq \sum_{j=-M}^{M} |a_{n+j}| \to 0$$

Since $\lim_{n\to\infty} S_n = \sum_{n=1}^{\infty} a_n$ exists,

$$\lim_{n\to\infty} S'_n = \lim_{n\to\infty} S_n$$

✓

Theorem 6.22 (Absolute convergence implies absolute convergence of any rearrangement). Suppose $\sum a_n$ converges absolutely, then all of its rearrangement $\sum a_{f(n)}$ converges absolutely, and

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{f(n)}$$

Proof. We first show $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |a_{f(n)}|$, we then show that $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{f(n)}$.

Step 1: $\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |a_{f(n)}|$,
Let S'_n denote the n-th partial sum of $\sum |a_{f(n)}|$. We have

$$S'_n \le \sum_{n=1}^{\infty} |a_n|$$

Since $\sum_{n=1}^{\infty} |a_{f(n)}|$ nonnegative, boundedness of partial sum implies $\sum |a_{f(n)}|$ converges. By symmetry,

$$\sum_{n=1}^{\infty} |a_n| = \sum_{n=1}^{\infty} |a_{f(n)}|$$

Step 2: now WTS $\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_{f(n)}$. Note that if the sequence is nonnegative then we are done. We restrict our attention to sequences with a positive and negative part. Define

$$a_n^+ = \begin{cases} a_n, \text{ if } a_n > 0 \\ 0 \text{ if } a_n \ge 0 \end{cases}$$

$$a_n^- = \begin{cases} -a_n, \text{ if } a_n < 0 \\ 0 \text{ if } a_n \ge 0 \end{cases}$$

Hence we have $\forall n$, $a_n^{\pm} \ge 0$. By comparison test, $\sum a_n^{\pm}$ converges. Now note that

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_n^+ - \sum_{n=1}^{\infty} a_n^-$$

Hence

$$\sum_{n=1}^{\infty} a_n = \sum_{n=1}^{\infty} a_n^+ - \sum_{n=1}^{\infty} a_n^- = \sum_{n=1}^{\infty} a_{f(n)} n^+ - \sum_{n=1}^{\infty} a_{f(n)} n^- = \sum_{n=1}^{\infty} a_{f(n)}$$

✓

Remark. For any series $\sum_{n=1}^{\infty} a_n$, write the n-th partial sum as S_n, then

$$S_n = S_n^+ - S_n^-$$

and if the limits of $S_n, S_n^{\pm}$ exist,

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = \lim_{n\to\infty} S_n^+ - \lim_{n\to\infty} S_n^-$$

Remark. add in remark

Theorem 6.23 (Riemann Rearrangement Theorem)**.**

Proof. add in proof ✓

Theorem 6.24 (Matrix product of absolutely convergent series)**.**

Proof. add in proof ✓

Theorem 6.25 (Cauchy product of convergent and absolutely convergent series)**.**

Proof. add in proof ✓

7 Appendix

7.1 Equivalent results in $\mathbb{R}$

Remark. In $\mathbb{R}$, TFAE

- Completeness Axiom (CA)
- Bolzano-Weierstrass (BW)
- Heine-Borel (HB)
- Nested Interval Property (NIP)
- Cauchy Convergence Criterion (CCC)
- Monotone Convergence Theorem (MCT)

This presents 30 potential proofs.

The table below links to each implication. The entry in row X, column Y is the proof of $X \Rightarrow Y$.

$\Rightarrow$	CA	BW	HB	NIP	CCC	MCT
CA	—	7.1.1	7.1.2	7.1.3	7.1.4	7.1.5
BW	7.1.6	—	7.1.7	7.1.8	7.1.9	7.1.10
HB	7.1.11	7.1.12	—	7.1.13	7.1.14	7.1.15
NIP	7.1.16	7.1.17	7.1.18	—	7.1.19	7.1.20
CCC	7.1.21	7.1.22	7.1.23	7.1.24	—	7.1.25
MCT	7.1.26	7.1.27	7.1.28	7.1.29	7.1.30	—

7.1.1 Completeness Axiom ⇒ Bolzano-Weierstrass

Proof. Exercise ✓

7.1.2 Completeness Axiom ⇒ Heine-Borel

Proof. Exercise ✓

7.1.3 Completeness Axiom ⇒ Nested Interval Property

Proof. Exercise ✓

7.1.4 Completeness Axiom ⇒ Cauchy Convergence Criterion

Proof. Exercise ✓

7.1.5 Completeness Axiom ⇒ Monotone Convergence Theorem

Proof. Exercise ✓

7.1.6 Bolzano-Weierstrass ⇒ Completeness Axiom

Proof. Exercise ✓

7.1.7 Bolzano-Weierstrass ⇒ Heine-Borel

Proof. Exercise ✓

7.1.8 Bolzano-Weierstrass ⇒ Nested Interval Property

Proof. Exercise ✓

7.1.9 Bolzano-Weierstrass ⇒ Cauchy Convergence Criterion

Proof. Exercise ✓

7.1.10 Bolzano-Weierstrass ⇒ Monotone Convergence Theorem

Proof. Exercise ✓

7.1.11 Heine-Borel ⇒ Completeness Axiom

Proof. Exercise ✓

7.1.12 Heine-Borel ⇒ Bolzano-Weierstrass

Proof. Exercise ✓

7.1.13 Heine-Borel ⇒ Nested Interval Property

Proof. Exercise ✓

7.1.14 Heine-Borel ⇒ Cauchy Convergence Criterion

Proof. Exercise ✓

7.1.15 Heine-Borel ⇒ Monotone Convergence Theorem

Proof. Exercise ✓

7.1.16 Nested Interval Property ⇒ Completeness Axiom

Proof. Exercise ✓

7.1.17 Nested Interval Property ⇒ Bolzano-Weierstrass

Proof. Exercise ✓

7.1.18 Nested Interval Property ⇒ Heine-Borel

Proof. Exercise ✓

7.1.19 Nested Interval Property ⇒ Cauchy Convergence Criterion

Proof. Exercise ✓

7.1.20 Nested Interval Property ⇒ Monotone Convergence Theorem

Proof. Exercise ✓

7.1.21 Cauchy Convergence Criterion ⇒ Completeness Axiom

Proof. Exercise ✓

7.1.22 Cauchy Convergence Criterion ⇒ Bolzano-Weierstrass

Proof. Exercise ✓

7.1.23 Cauchy Convergence Criterion ⇒ Heine-Borel

Proof. Exercise ✓

7.1.24 Cauchy Convergence Criterion ⇒ Nested Interval Property

Proof. Exercise ✓

7.1.25 Cauchy Convergence Criterion ⇒ Monotone Convergence Theorem

Proof. Exercise ✓

7.1.26 Monotone Convergence Theorem ⇒ Completeness Axiom

Proof. Exercise ✓

7.1.27 Monotone Convergence Theorem ⇒ Bolzano-Weierstrass

Proof. Exercise ✓

7.1.28 Monotone Convergence Theorem ⇒ Heine-Borel

Proof. Exercise ✓

7.1.29 Monotone Convergence Theorem ⇒ Nested Interval Property

Proof. Exercise ✓

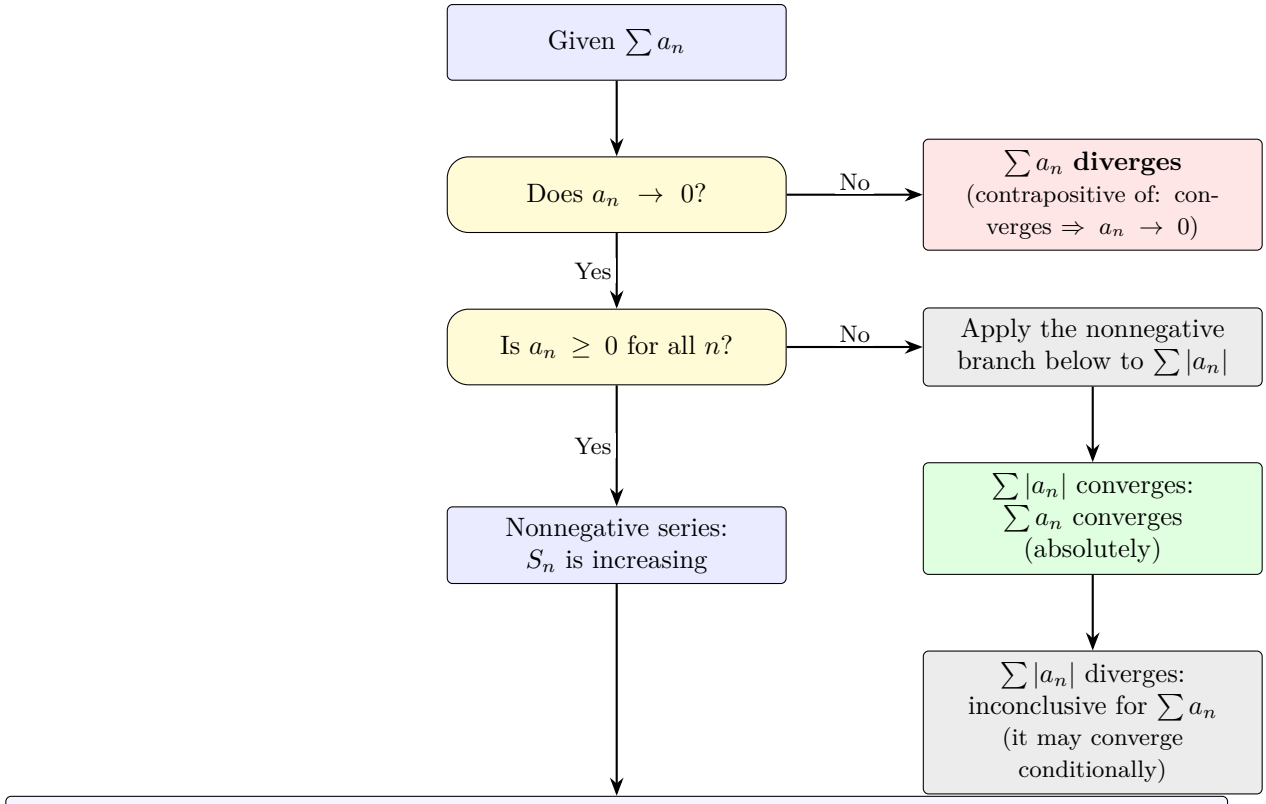
7.1.30 Monotone Convergence Theorem ⇒ Cauchy Convergence Criterion

Proof. Exercise ✓

7.2 Series encountered and their convergence properties

7.2.1 Flow chart: which result applies to $\sum a_n$?

Remark (Flow chart).

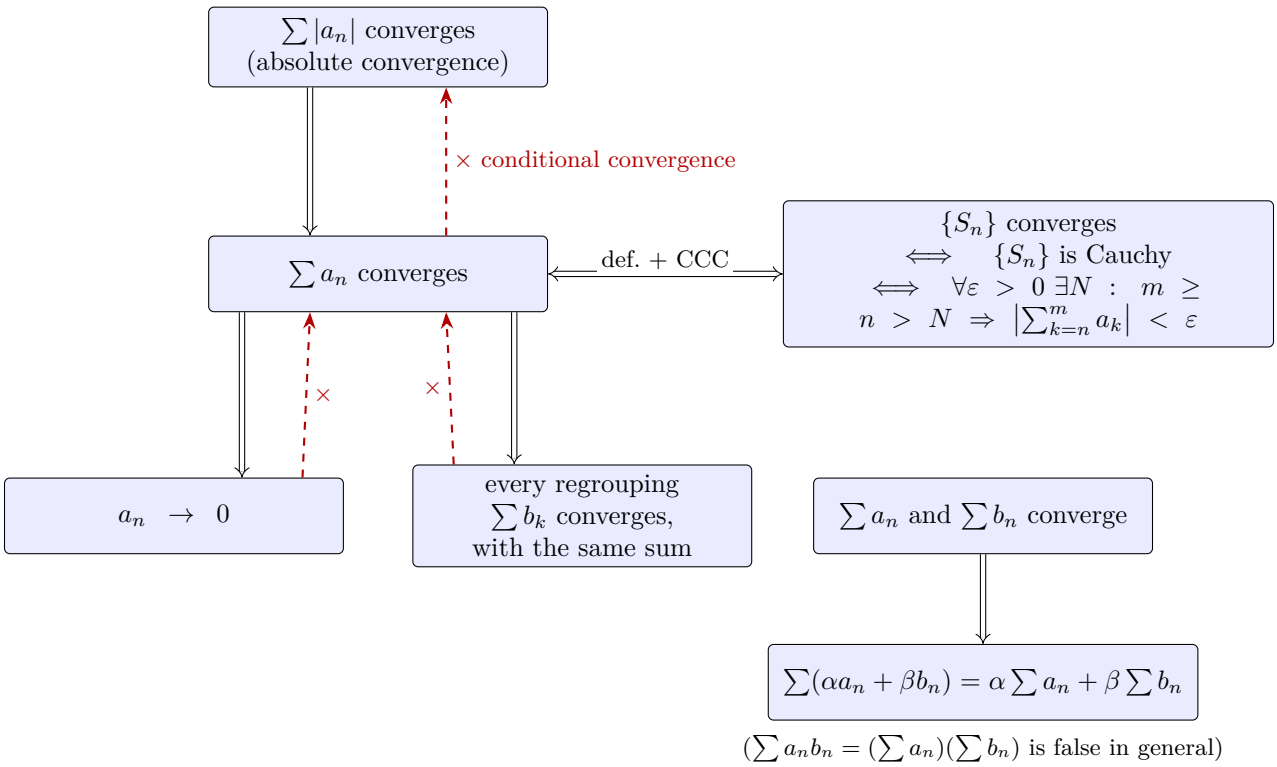


Tool	**Converges if**	**Diverges if**
Partial sums	S_n bounded (or has a bounded subsequence)	S_n unbounded ($S_n \to \infty$)
Comparison	$a_n \leq b_n$ eventually, $\sum b_n$ converges	$a_n \geq b_n \geq 0$ eventually, $\sum b_n$ diverges
Ratio ($a_n \neq 0$)	$\limsup \frac{a_{n+1}}{a_n} < 1$	$\liminf \frac{a_{n+1}}{a_n} > 1$
Root	$\limsup \sqrt[n]{a_n} < 1$	$\limsup \sqrt[n]{a_n} > 1$
Benchmarks	$\sum a^{n-1}$ for $\lvert a\rvert < 1$; $\sum \frac{1}{n^p}$ for $p > 1$	$\sum a^{n-1}$ for $\lvert a\rvert \geq 1$; $\sum \frac{1}{n^p}$ for $p \leq 1$

For the ratio and root tests, a limsup or liminf equal to 1 is inconclusive.

7.2.2 Implication map

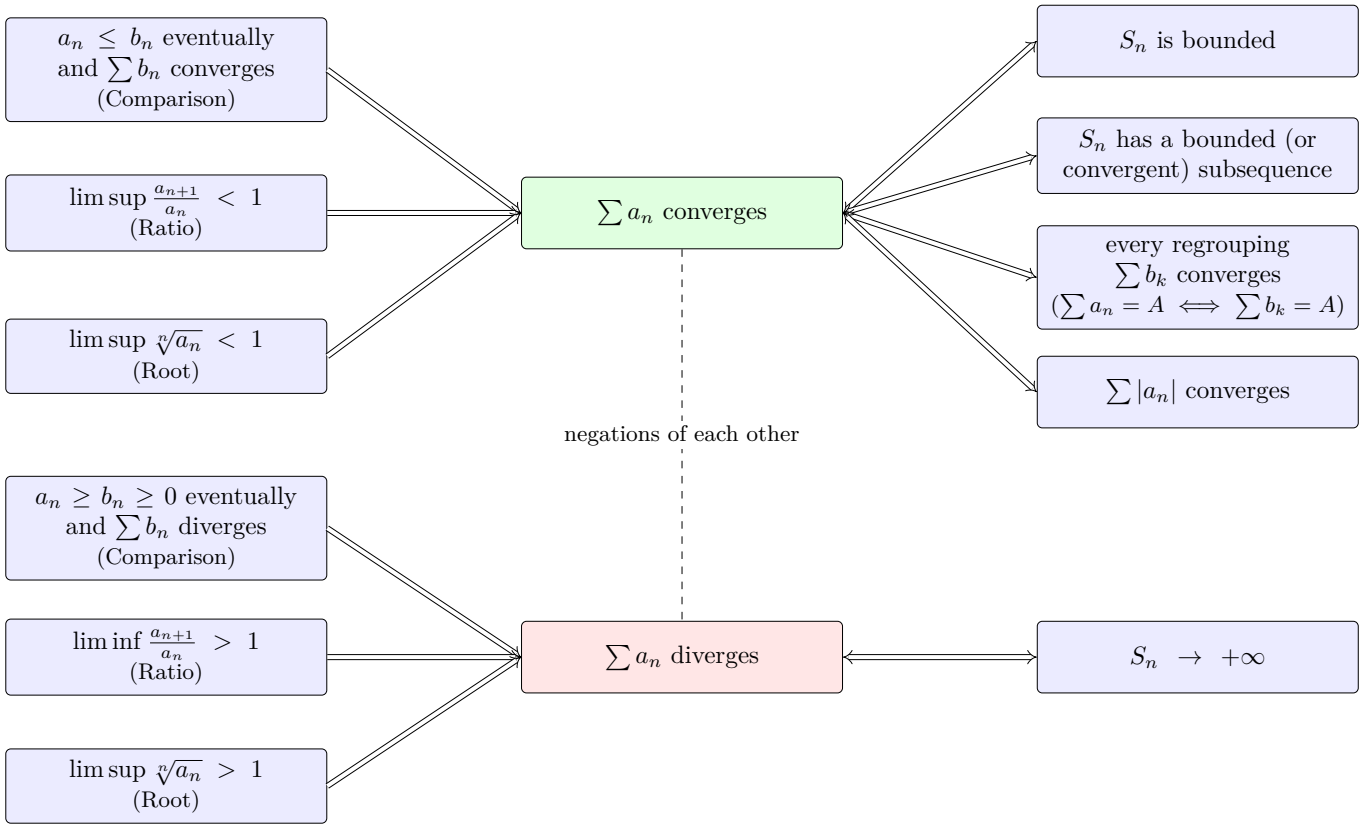
Remark (Implications for general series)**.** Here $\{a_n\} \subset \mathbb{R}$ has no sign condition. A double arrow is an implication. A dashed red arrow marked $\times$ is a converse that is *false*.



The converse failures are: the harmonic series ($a_n \to 0$ but diverges) and $\sum(-1)^n$ (regroups to $\sum 0 = 0$, but diverges).

Absolute convergence is proved from the Cauchy criterion and the triangle inequality.

Remark (Implications for nonnegative series)**.** Here $a_n \geq 0$, so $\{S_n\}$ is increasing and the monotone convergence theorem applies.



Benchmarks. Geometric: $\sum a^{n-1}$ converges (to $\frac{1}{1-a}$) iff $|a| < 1$. p-series: $\sum \frac{1}{n^p}$ converges iff $p > 1$ (harmonic series is $p = 1$). For the ratio and root tests, a limsup or liminf equal to 1 is inconclusive.